Introduction to Partial Differential Equations

Qi Zhou

Lanzhou University

Email:zhouq2020@lzu.edu.cn

Abstract

Notes for 2020 BICMR Summer School for Differential Geometry.This course is taught by Jianchun Chu from Northwestern University.The main reference is Elliptic partial differential equations of second order by D.Gilbarg and N.S.Trudinger.

Contents

1 Poisson Equations **1**

1.1 Sup-norms of solutions . . . 1

1.2 Hölder spaces . . . 5

1.3 $C^{2,\alpha}$ estimate of solutions . . . 6

1.4 Dirichlet problem . . . 13

2 Schauder Theory—Non-divergence Form **17**

2.1 Interior $C^{2,\alpha}$ Schauder estimates . . . 17

2.2 Schauder estimates in weighted Hölder norms . . . 27

2.3 The Dirichlet problem–methods of continuity(balls) . . . 30

2.4 The Dirichlet problem–Perron's method(general bounded domain) . . . 33

1 Poisson Equations

1.1 Sup-norms of solutions

Let Ω be a domain in $\mathbb{R}^n$ and f be a continuous function in Ω.

Definition 1.1 (Poisson Equation) *Assume $u \in C^2(\Omega)$,the **Poisson Equation** has the form*

$$\Delta u = f$$

in Ω.

Lemma 1.1 *Let Ω be a bounded domain in $\mathbb{R}^n$.Then for any $f \in C(\Omega)$ and $\varphi \in C(\partial\Omega)$,there exists at most one solution $u \in C^2(\Omega) \cap C(\overline{\Omega})$ of the problem*

$$\begin{cases} \Delta u = f & in \quad \Omega \\ u = \varphi & on \quad \partial\Omega \end{cases}$$

Proof It is trivial. □

We derive an a priori estimate for solutions of the Dirichlet problem.

Theorem 1.1 *Let Ω be a bounded domain in $\mathbb{R}^n$,f be a continuous function in $\overline{\Omega}$ and φ be a continuous function on $\partial\Omega$.Suppose $u \in C^2(\Omega) \cap C(\overline{\Omega})$ satisfies*

$$\begin{cases} \Delta u = f & in \quad \Omega \\ u = \varphi & on \quad \partial\Omega \end{cases}$$

Then

$$\sup_{\Omega} |u| \leq \sup_{\partial\Omega} |\varphi| + C \sup_{\Omega} |f|$$

where C is a positive constant depending only on n and $\mathrm{diam}(\Omega)$.

Proof Set

$$F = \sup_{\Omega} |f|, \quad \Phi = \sup_{\partial\Omega} |\varphi|$$

Then

$$\begin{cases} \Delta(\pm u) = \pm f \geq -F & in \quad \Omega \\ \pm u = \pm\varphi & on \quad \partial\Omega \end{cases}$$

Without loss of generally,we assume $\Omega \subset B_R$,for $R > 0$.Set

$$v(x) = \Phi + \frac{F}{2n}(R^2 - |x|^2) \quad \forall x \in \Omega$$

Then $\Delta v = -F$ in Ω and $v \geq \Phi$ on $\partial\Omega$.Hence v satisfies

$$\begin{cases} \Delta v \leq -F & in \quad \Omega \\ v \geq \Phi & on \quad \partial\Omega \end{cases}$$

Therefore,

$$\begin{cases} \Delta(\pm u - v) \geq 0 & in \quad \Omega \\ \pm u - v \leq 0 & on \quad \partial\Omega \end{cases}$$

We obtain $\pm u - v \leq 0$,and hence $|u| \leq v$ in Ω.Therefore,

$$|u(x)| \leq \Phi + \frac{1}{2n}(R^2 - |x|^2)F \quad \forall x \in \Omega$$

This yields the desired result. □

Under extra assumptions on boundary values,we can improve estimates on solutions.We will do this for the Dirichlet problem in balls with zero boundary value.For any ball B in $\mathbb{R}^n$,we write

$$d_x = \mathrm{dist}(x, \partial B)$$

Theorem 1.2 *Let B be a ball in $\mathbb{R}^n$ and $f \in C(B)$ satisfy*

$$\sup_{x \in B} d_x^{2-\beta} |f(x)| < \infty$$

for some $\beta \in (0,1)$.Suppose $u \in C(\overline{B}) \cap C^2(B)$ is a solution of

$$\begin{cases} \Delta u = f & in \quad B \\ u = 0 & on \quad \partial B \end{cases}$$

Then

$$\sup_{x \in B} d_x^{-\beta} |u(x)| \leq C \sup_{x \in B} d_x^{2-\beta} |f(x)|$$

where C is a positive constant depending only on n and β.

Proof We assume $B = B_R$ and

$$N = \sup_{x \in B} d_x^{2-\beta} |f(x)| < \infty$$

Set

$$v(x) = (R^2 - |x|^2)^\beta$$

A straightforward calculation yields

$$D_i v = -2\beta (R^2 - |x|^2)^{\beta-1} x_i$$

$$D_{ii} v = -2\beta (R^2 - |x|^2)^{\beta-1} + \beta(\beta-1)(R^2 - |x|^2)^{\beta-2}(-2x_i)^2$$

then

$$\Delta v(x) = -2\beta (R^2 - |x|^2)^{\beta-2} (n(R^2 - |x|^2) + 2(1-\beta)|x|^2)$$

For $|x| < R$,

$$n(R^2 - |x|^2) + 2(1-\beta)|x|^2 = (n + 2(\beta-1))(R^2 - |x|^2) + 2(1-\beta)R^2 \geq 2(1-\beta)R^2$$

by $n - 2(1-\beta) > 0$.Hence,for any $|x| < R$,

$$\begin{aligned} \Delta v(x) &\leq -4\beta(1-\beta)R^2 (R^2 - |x|^2)^{\beta-2} \\ &= -4\beta(1-\beta) \left(\frac{R}{R+|x|} \right)^2 (R - |x|)^{\beta-2} (R + |x|)^\beta \\ &\leq -4\beta(1-\beta) \frac{R^2}{R^2 + 2R|x| + |x|^2} \cdot R^\beta (R - |x|)^{\beta-2} \\ &\leq -4\beta(1-\beta) \frac{R^2}{4R^2} \cdot R^\beta (R - |x|)^{\beta-2} \\ &= -\beta(1-\beta) R^3 (R - |x|)^{\beta-2} \end{aligned}$$

Note $d_x = R - |x|$ for any $x \in B$.Then,

$$\Delta(\pm u) = \pm f \geq -N d_x^{\beta-2} = -N(R - |x|)^{\beta-2} \geq \frac{N \Delta v}{\beta(1-\beta)R^\beta}$$

Hence,

$$\begin{cases} \Delta(\pm u - (\beta(1-\beta)R^\beta)^{-1} N v) \geq 0 & in \quad B \\ \pm u - (\beta(1-\beta)R^\beta)^{-1} N v \leq 0 & on \quad \partial B \end{cases}$$

We have

$$\pm u - \frac{N v}{\beta(1-\beta)R^\beta} \leq 0$$

in B.Therefore,

$$\begin{aligned}|u(x)| &\leq \frac{Nv(x)}{\beta(1-\beta)R^\beta} = \frac{N(R^2-|x|^2)^\beta}{\beta(1-\beta)R^\beta} \\ &\leq \frac{N(R-|x|)^\beta(R+|x|)^\beta}{\beta(1-\beta)R^\beta} \leq \frac{2^\beta}{\beta(1-\beta)}N(R-|x|)^\beta \\ &= \frac{2^\beta}{\beta(1-\beta)}Nd_x^\beta\end{aligned}$$

for any $x \in B$. □

We now make a comment on **Theorem1.1** and **Theorem1.2**.Estimates in these results are what we call the a **priori estimates**.Namely,we assume solutions already exist and then derive certain norms of solutions in terms of data in the problem.It is important to note that establishing a priori estimates is in fact the first step in proving the existence of solutions.We will deal with the a priori estimates in the most of this book.A closely related issue here the regularity of solutions such as continuity and differentiability.Solutions of a particular PDE can only be obtained if the right kind of regularity,or the right kind of norms,are employed.

Now we study more closely the Poisson equation $\Delta u = f$,which means that a certain combination of the second derivatives of u is determined by the given function f,the nonhomogeneous term.If u is C^2,then obviously f is continuous.It is natural to ask whether the Poisson equations admit any C^2 solution if the nonhomogeneous terms are continuous terms are continuous.In terms of a priori estimates,this is equivalent to the question whether the sup-norms of nonhomogeneous terms can control the second derivatives of solutions.The answer turns out to be negative.

There exists a continuous function f such that $\Delta u = f$ does not admit any C^2 solution.

EXAMPLE 1.1 *Let f be a function in $B_1 \subset \mathbb{R}^2$ defined by $f(0) = 0$ and*

$$f(x) = \frac{x_2^2 - x_1^2}{|x|^2}\left(\frac{2}{(-\log|x|)^{1/2}} + \frac{1}{4(-\log|x|)^{3/2}}\right)$$

for any $x \in B_1 \setminus \{0\}$.Then f is continuous in B_1.Consider

$$\Delta u = f \quad in \quad B_1$$

Define u in B_1 by $u(0) = 0$ and

$$u(x) = (x_1^2 - x_2^2)(-\log|x|)^{1/2}$$

for any $x \in B_1 \setminus \{0\}$.Then $u \in C(B_1) \cap C^\infty(B_1 \setminus \{0\})$.A straightforward calculation shows that u satisfies $\Delta u = f$ in $B_1 \setminus \{0\}$ and

$$\lim_{x\to 0} u_{x_1x_1}(x) = 0$$

Hence,u is not in $C^2(B_1)$.

Next,we prove that $\Delta u = f$ in B_1 has no C^2 solutions.

*Suppose,to the contrary,that there exists a C^2 solution v of $\Delta u = f$ in B_1.For a fixed $R \in (0,1)$,the function $w = u - v$ is harmonic in $B_R \setminus \{0\}$.Now $u \in C(\overline{B}_R)$ and $v \in C^2(B_R)$,so $w \in C(B_R)$.Thus w is continuous at the origin.By **Theorem??**,w is harmonic in B_R and therefore belongs to $C^2(B_R)$.In particular,u is C^2 at the origin,which is a contradiction.*

Example illustrates that the C^2-spaces,or any C^k-spaces,are not adapted to the Poisson equation.A further investigation reveals that solutions in this example fail to be C^2 because the modulus of continuity of f does not decay to zero fast enough.If there is a better assumption than the continuity of f,then the modulus of

continuity of D^2u can be estimated.Better adapted to the Poisson equation,are **Hölder spaces**.

1.2 Hölder spaces

In this section,we introduce Hölder spaces.
Let Ω be a bounded domain in $\mathbb{R}^n$.

Definition 1.2 (Locally Hölder Continuous) *We denote by $C^\alpha(\Omega)$ the collection of all continuous functions f defined in Ω such that,for any $\Omega' \subset\subset \Omega$,*

$$|f(x) - f(y)| \leq C|x - y|^\alpha \quad \forall x, y \in \Omega'$$

*where C is constant depending on f and Ω'.Functions in $C^\alpha(\Omega)$ are called **locally Hölder continuous functions in Ω with exponent** $\alpha \in (0, 1)$.*

Definition 1.3 (Hölder Continuous) *We denote by $C^\alpha(\overline{\Omega})$ the collection of all continuous functions f in $\overline{\Omega}$ such that*

$$|f(x) - f(y)| \leq C|x - y|^\alpha \quad \forall x, y \in \overline{\Omega}$$

*where C is constant depending on f.Functions in $C^\alpha(\overline{\Omega})$ are called **Hölder continuous functions in Ω with exponent** $\alpha \in (0, 1)$.*

We now introduce Hölder differentiable functions.This is easy to do for functions defined in open sets.For each nonnegative integer k,we define

$$C^{k,\alpha}(\Omega) = \{f \in C^k(\Omega) \mid D^k f \in C^\alpha(\Omega)\}$$

It is a bit tedious when functions are defined in closure of open sets since we need to take care of derivatives restricted to boundary.We first discuss a special case.For any $r > 0$,set

$$B_r^+ = \{x = (x_1, x_2, \cdots, x_{n-1}, x_n) \in B_r \mid x_n > 0\}$$

$$\Sigma_r = \{x = (x_1, x_2, \cdots, x_{n-1}, x_n) \in B_r \mid x_n = 0\}$$

For a fixed $R > 0$,let f be a continuous function in $B_R^+ \cup \Sigma_R$.We can define first derivatives of f in $B_R^+ \cup \Sigma_R$. In fact,x_i-derivatives of f can be defined as usual,for $i = 1, \cdots, n-1$ in $B_R^+ \cup \Sigma_R$.The x_n-derivative along Σ_R is understood as a one-side derivatives.In general,we can define C^k-functions and $C^{k,\alpha}$-functions in $B_R^+ \cup \Sigma_R$.

Definition 1.4 ($C^{k,\alpha}$-boundary) *Let Ω be a bounded domain in $\mathbb{R}^n$.The boundary $\partial\Omega$ is called $C^{k,\alpha}$ if,for any $x_0 \in \partial\Omega$,there exist a neighborhood U_{x_0} of x_0,a neighborhood V_{x_0} of the origin in $\mathbb{R}^n$ and a map Φ_{x_0} : $U_{x_0} \longrightarrow V_{x_0}$ such that*
(1)$\phi_{x_0} : U_{x_0} \longrightarrow V_{x_0}$ is one-to-one and onto.
(2)Φ_{x_0} and $\Phi_{x_0}^{-1}$ are $C^{k,\alpha}$.
(3)$\Phi_{x_0}(U_{x_0} \cap \Omega) = V_{x_0} \cap \{x \mid x_n > 0\}$.
The pair (U_{x_0}, Φ_{x_0}) is called a coordinate patch of $\partial\Omega$ around x_0.

Now the regularity of u near boundary can defined by that of $u \circ \Phi_{x_0}^{-1}$ with the help of coordinate patches.
Let Ω be a bounded domain in $\mathbb{R}^n$ with a $C^{k,\alpha}$-boundary.The space $C^{k,\alpha}(\overline{\Omega})$ is the collection of functions f defined in $\overline{\Omega}$ such that $f \in C^{k,\alpha}(\Omega)$ and,for any $x_0 \in \partial\Omega$ and any coordinate patch (U_{x_0}, Φ_{x_0}),the function $u \circ \Phi_{x_0}^{-1}$ is $C^{k,\alpha}$ in $\overline{\Phi_{x_0}(U_{x_0})} \cap B_r$ for some $B_r \subset\subset \Phi_{x_0}(U_{x_0})$.
When discussing regularity of solutions up to boundaries,we also need spaces $C^{k,\alpha}(\Omega \cap \Sigma)$ for a boundary portion $\Sigma \subset \partial\Omega$.These spaces can be defined accordingly.We note that $C^{k,\alpha}(\overline{\Omega}) = C^{k,\alpha}(\Omega \cup \partial\Omega)$.

Definition 1.5 (Hölder Space) *For each positive integer k,we define*

$$C^{k,\alpha}(\overline{\Omega}) = \{f \in C^k(\overline{\Omega}) \mid D^k f \in C^\alpha(\overline{\Omega})\}$$

For unification,we sometimes write $C^{0,\alpha}(\overline{\Omega}) = C^\alpha(\overline{\Omega})$.For each nonnegative integer k,$C^{k,\alpha}(\overline{\Omega})$ is called a ***Hölder space****.*

We now introduce norms to make Hölder spaces complete.

Definition 1.6 (Hölder Semi-norm) *For each $f \in C^\alpha(\overline{\Omega})$,we define*

$$[f]_{C^\alpha(\overline{\Omega})} = \sup_{x,y\in\overline{\Omega},x\neq y} \frac{|f(x)-f(y)|}{|x-y|^\alpha}$$

which is called the ***Hölder semi-norm*** *of f in $\overline{\Omega}$.*

Definition 1.7 (Hölder Norm) *For each $u \in C^k(\overline{\Omega})$ and $C^{k,\alpha}(\overline{\Omega})$,we define*

$$|u|_{C^k(\overline{\Omega})} = \sum_{i=0}^{k} \sup_{\overline{\Omega}} |D^i u|$$

$$|u|_{C^{k,\alpha}(\overline{\Omega})} = |u|_{C^k(\overline{\Omega})} + [D^k u]_{C^\alpha(\overline{\Omega})}$$

which is called the ***Hölder norm*** *of u in $\overline{\Omega}$.*

1.3 $C^{2,\alpha}$ estimate of solutions

In this section,we study regularity of solutions of the Poisson equation.
We first study the regularity of a special solution of the Poisson equations.

Definition 1.8 (Newtonian Potential) *If Ω is a bounded domain and f is a bounded function,we define*

$$w(x) = \int_\Omega \Gamma(x-y)f(y)dy$$

where Γ is the fundamental solution that

$$\Gamma(x-y) = \begin{cases} \dfrac{1}{2\pi}\log|x-y| & n=2 \\ \dfrac{1}{(2-n)\omega_n}|x-y|^{2-n} & n\geq 3 \end{cases}$$

This is called the ***Newtonian potential*** *of f in Ω.*

Remark 1.1 *We will write $w_{f,\Omega}$ to emphasize the dependence on f and the domain Ω.*
It is easy to see that w is a well-defined continuous function in $\mathbb{R}^n$,although Γ has a regularity.

Lemma 1.2 *Let $\Omega = B_1$ be the unit ball in $\mathbb{R}^n$,f be a bounded function in B_1 and w be the Newtonian potential of f in B_1.Then,$w \in C^1(\mathbb{R}^n)$ and,for any $x \in \mathbb{R}^n$ and $i = 1,\cdots,n$,*

$$D_i w(x) = \int_{B_1} D_{x_i}\Gamma(x-y)f(y)dy$$

Moreover,

$$|w|_{C^1(\overline{B}_1)} \leq C|f|_{L^\infty(B_1)}$$

where C is a positive constant depending only on n.

Proof Set

$$v_i(x) = \int_{B_1} D_{x_i}\Gamma(x-y)f(y)dy$$

Since

$$|D\Gamma(x-y)| = \begin{cases} O(|x-y|^{-1}) & n=2 \\ O(|x-y|^{1-n}) & n \geq 3 \end{cases}$$

and $f(y)$ is bounded function in B_1,then $v_i(x)$ is a well-defined function in $\mathbb{R}^n$ and is continuous in $\mathbb{R}^n$.
Fix a function $\eta \in C^1(\mathbb{R})$ with $0 \leq \eta \leq 1$,$0 \leq \eta' \leq 2$,$\eta(t) = 0$ for $t \leq 1$ and $\eta(t) = 1$ for $t \geq 2$.For any $\varepsilon > 0$,we define

$$w_\varepsilon(x) = \int_{B_1} \Gamma\eta_\varepsilon f(y)dy$$

where we write $\Gamma = \Gamma(x-y)$ and $\eta_\varepsilon = \eta\left(\frac{|x-y|}{\varepsilon}\right)$.Then $\Gamma(x-y)\cdot\eta_\varepsilon(|x-y|) \in C^1(\mathbb{R}^n)$,hence $w_\varepsilon(x) \in C^1(\mathbb{R}^n)$.
We will prove that $w_\varepsilon(x)$ converge uniformly in compact subsets of $\mathbb{R}^n$ to $w(x)$,

$$\begin{aligned} |w(x) - w_\varepsilon(x)| &= \left|\int_{B_1} \Gamma(|x-y|)(1-\eta_\varepsilon(|x-y|))f(y)dy\right| \\ &\leq \int_{|x-y|\leq 2\varepsilon} |\Gamma(|x-y|)| \cdot (1-\eta_\varepsilon) \cdot |f(y)|dy \\ &\leq \int_{|x-y|\leq 2\varepsilon} |\Gamma(|x-y|)dy \cdot \sup|f| \\ &= \begin{cases} \int_0^{2\varepsilon} -\log r \cdot rdr \cdot \sup|f| \leq -\log 2\varepsilon \cdot (2\varepsilon)^2 \cdot \sup|f| & n=2 \\ \int_0^{2\varepsilon} -\frac{1}{2-n} r^{2-n} \cdot r^{n-1}dr \cdot \sup|f| \leq (2\varepsilon)^2 \cdot \sup|f| & n \geq 3 \end{cases} \end{aligned}$$

where we apply the fact that the integrand is zero for $|x-y| \geq 2\varepsilon$,and the last equality by

$$\int_0^{2\varepsilon} \omega_n r^{n-1}dr = \int_{B_{2\varepsilon}(x)} u(x-y)dy.$$

Since $w_\varepsilon(x) \in C^1(\mathbb{R}^n)$,then $w \in C^1(\mathbb{R}^n)$.
We will prove that $D_i w_\varepsilon(x)$ converge uniformly in compact subsets of $\mathbb{R}^n$ to $v_i(x)$,

$$\begin{aligned} |v_i(x) - D_i w_\varepsilon(x)| &= \left|\int_{B_1} D_i((1-\eta_\varepsilon)\Gamma(|x-y|))f(y)dy\right| \\ &\leq \int_{|x-y|\leq 2\varepsilon} |D_i((1-\eta_\varepsilon)\Gamma(|x-y|))|\, dy \cdot \sup|f| \\ &\leq \int_{|x-y|\leq 2\varepsilon} (|D_i\Gamma| + \frac{2}{\varepsilon}|\Gamma|)dy \cdot \sup|f| \\ &\leq \begin{cases} \int_0^{2\varepsilon} \left(\frac{1}{r} + (-\log r)\cdot\frac{2}{\varepsilon}\right) rdr \cdot \sup|f| & n=2 \\ \int_0^{2\varepsilon} \left(r^{1-n} + \frac{2}{\varepsilon}r^{2-n}\right) \cdot r^{n-1}dr \cdot \sup|f| & n \geq 3 \end{cases} \\ &\leq \begin{cases} 4\varepsilon(1+|\log(2\varepsilon)|) \cdot \sup|f| & n=2 \\ 6\varepsilon \cdot \sup|f| & n \geq 3 \end{cases} \end{aligned}$$

Hence $D_i w = v_i$. □

Remark 1.2 *(1)If w is C^2 and $\Delta w = f$ in B_1,then any solution of $\Delta u = f$ differs from w by an addition of*

a harmonic function.
(2)Since harmonic functions are smooth,regularity of arbitrary solutions of $\Delta u = f$ is determined by that Newtonian potential w.
(3)We cannot differentiate $D_i w$ to get second derivatives of w if f is simply bounded since $D^2\Gamma$ is not integrable.We need extra assumptions on f.

Lemma 1.3 *Let $\Omega = B_1$ be the unit ball in $\mathbb{R}^n$,$f \in L^\infty(B_1) \cap C^\alpha(B_1)$ for some $\alpha \in (0,1)$ and w be the Newtonian potential of f in B_1.Then $w \in C(\overline{B}_1) \cap C^{2,\alpha}(B_1)$ and $\Delta w = f$ in B_1.Moreover*

$$|w|_{C^{2,\alpha}(B_{1/2})} \leq C|f|_{C^\alpha(B_1)}$$

where C is a positive constant depending only on n and α.

Proof We first prove $w \in C^2(B_1)$ and,for any $x \in B_1$ and $i,j = 1,2,\cdots,n$,

$$D_{ij}w(x) = \int_{B_1} D_{x_ix_j}\Gamma(x-y)(f(y)-f(x))dy - f(x)\int_{\partial B_1} D_{x_j}\Gamma(x-y)\nu_i(y)dS_y$$

We set

$$v_{ij}(x) = \int_{B_1} D_{x_ix_j}\Gamma(x-y)(f(y)-f(x))dy - f(x)\int_{\partial B_1} D_{x_j}\Gamma(x-y)\nu_i(y)dS_y$$

Since $|D\Gamma| = O(|x-y|^{1-n})$,$|D^2\Gamma| = O(|x-y|^{-n})$ and $|f(y)-f(x)| = O(|x-y|^\alpha)$,hence $v_{ij}(x)$ is a well defined function in B_1.For any $\varepsilon > 0$,define

$$v_{i,\varepsilon}(x) = \int_{B_1} D_i\Gamma\eta_\varepsilon f(y)dy$$

where $\eta_\varepsilon = \eta\left(\frac{|x-y|}{\varepsilon}\right)$,$0 \leq \eta \leq 1$,$0 \leq \eta' \leq 2$,$\eta = \begin{cases} 0 & t \leq 1 \\ 1 & t \geq 2 \end{cases}$.
Since $D_i\Gamma\eta_\varepsilon \in C^1$,then $v_{i,\varepsilon} \in C^1(B_1)$ and,by differentiating,we obtain

$$\begin{aligned}
D_j v_{i,\varepsilon}(x) &= \int_{B_1} D_j(D_i\Gamma\eta_\varepsilon)f(y)dy \\
&= \int_{B_1} D_j(D_i\Gamma\eta_\varepsilon)(f(y)-f(x))dy + f(x)\int_{B_1} D_j(D_i\Gamma\eta_\varepsilon)dy \\
&= \int_{B_1} D_j(D_i\Gamma\eta_\varepsilon)(f(y)-f(x))dy - f(x)\int_{B_1} D_{y_j}(D_i\Gamma\eta_\varepsilon)dy \\
&= \int_{B_1} D_j(D_i\Gamma\eta_\varepsilon)(f(y)-f(x))dy - f(x)\int_{\partial B_1} D_i\Gamma\eta_\varepsilon\nu_j dS_y \\
&= \int_{B_1} D_j(D_i\Gamma\eta_\varepsilon)(f(y)-f(x))dy - f(x)\int_{\partial B_1} D_i\Gamma\nu_j dS_y
\end{aligned}$$

for ε sufficiently small.Hence

$$\begin{aligned}
|v_{ij}(x) - D_j v_{i,\varepsilon}(x)| &\leq \int_{|y-x|\leq 2\varepsilon} |D_j((1-\eta_\varepsilon)D_i\Gamma)(f(y)-f(x))|dy \\
&\leq [f]_{\alpha,x}\int_{|y-x|\leq 2\varepsilon}\left(|D_{ij}\Gamma| + \frac{2}{\varepsilon}|D_i\Gamma|\right)\cdot|y-x|^\alpha dy \\
&\leq C\cdot[f]_{\alpha,x}\int_0^{2\varepsilon}\left(r^{-n} + \frac{2}{\varepsilon}r^{1-n}\right)r^\alpha\cdot r^{n-1}dr \\
&\leq C\cdot[f]_{\alpha,x}\cdot\left(\frac{1}{\alpha}(2\varepsilon)^\alpha + \frac{2}{\varepsilon}\cdot\frac{1}{\alpha+1}\cdot(2\varepsilon)^{\alpha+1}\right)
\end{aligned}$$

$$= C \cdot [f]_{\alpha,x} \cdot (2\varepsilon)^{\alpha} \left(\frac{1}{\alpha} + \frac{4}{\alpha+1} \right)$$

provided $|x| < 1 - 2\varepsilon$.Therefore,$D_j v_{i,\varepsilon}$ converges to v_{ij} uniformly in compact subsets of B_1 as $\varepsilon \longrightarrow 0$.Since $v_{i,\varepsilon}$ converges uniformly to $D_i w$ in B_1 as $\varepsilon \longrightarrow 0$,we obtain $w \in C^2(B_1)$ and $v_{ij} = D_{ij} w$.
Since

$$D_{ij} w(x) = \int_{B_1} D_{x_i x_j} \Gamma(x-y)(f(y) - f(x))dy - f(x) \int_{\partial B_1} D_{x_j} \Gamma(x-y) \nu_i(y) dS_y$$

by taking $i = j$,a summation yields

$$\Delta w(x) = \int_{B_1} \Delta_x \Gamma(x-y)(f(y) - f(x))dy + f(x) \int_{\partial B_1} \frac{\partial \Gamma}{\partial \nu_y}(x-y) dS_y$$

By $\Delta_x \Gamma(x-y) = 0$ for any $y \neq x$ and $\displaystyle\int_{\partial B_1} \frac{\partial \Gamma}{\partial \nu_y}(x-y) dS_y = 1$,we obtain

$$\Delta w(x) = f(x)$$

Next we prove $w \in C^{2,\alpha}(B_1)$.
For any $x \in B_{1/2}$,we have

$$D_{ij} w(x) = \int_{B_1} D_{ij} \Gamma(x-y)(f(y) - f(x))dy - f(x) \int_{\partial B_1} D_i \Gamma(x-y) \nu_j(y) dS_y$$

Hence,

$$\begin{aligned}
|D_{ij}(x)| &\leq C \cdot \left([f]_{\alpha} \int_{B_1} |x-y|^{\alpha-n} dy + |f| \int_{\partial B_1} |x-y|^{1-n} dS_y \right) \\
&\leq C \cdot \left([f]_{\alpha} \int_0^{3/2} r^{\alpha-n} \cdot \omega_n r^{n-1} dr + |f| \int_{\partial B_1} \left(\frac{1}{2}\right)^{1-n} dS_y \right) \\
&= C \cdot \left([f]_{\alpha} \omega_n \cdot \frac{1}{\alpha} \cdot \left(\frac{3}{2}\right)^{\alpha} + \left(\frac{1}{2}\right)^{1-n} \cdot |\partial B_1| \cdot |f| \right) \\
&\leq C \cdot (|f(x)| + [f]_{\alpha})
\end{aligned}$$

since $|x-y| \geq \dfrac{1}{2}$ for any $x \in B_{1/2}$ and $y \in \partial B_1$.
Now we will illustrate that we take any other $\overline{x} \in B_{1/2}$ and set $r = |x - \overline{x}|$ and obtain

$$\frac{|D_{ij} w(x) - D_{ij} w(\overline{x})|}{|x - \overline{x}|^{\alpha}} \leq C$$

From the above assumptions,we have

$$D_{ij} w(x) = \int_{B_1} D_{ij} \Gamma(x-y)(f(y) - f(x))dy - f(x) \int_{\partial B_1} D_i \Gamma(x-y) \nu_j(y) dS_y$$

and

$$D_{ij} w(\bar{x}) = \int_{B_1} D_{ij} \Gamma(\overline{x}-y)(f(y) - f(\overline{x}))dy - f(\overline{x}) \int_{\partial B_1} D_i \Gamma(\overline{x}-y) \nu_j(y) dS_y$$

If $r \geq \frac{1}{4}$.We have

$$\frac{|D_{ij}w(x) - D_{ij}w(\overline{x})|}{|x-\overline{x}|^\alpha} \leq \frac{1}{\left(\frac{1}{4}\right)^\alpha}(|D_{ij}w(x)| + |D_{ij}w(\overline{x})|) \leq C(|f(x)| + [f]_\alpha + |f(\overline{x})| + [f]_\alpha)$$

If $r < \frac{1}{4}$.Set $\hat{x} = \frac{x+\overline{x}}{2}$.Then

$$D_{ij}w(x) - D_{ij}(w(\overline{x})) = I_1 + I_2 + I_3 + I_4 + I_5 + I_6$$

where

$$\begin{aligned}
I_1 &= f(x)\int_{\partial B_1}(D_i\Gamma(x-y) - D_i\Gamma(\bar{x}-y))\nu_j(y)dS_y \\
I_2 &= (f(x) - f(\bar{x}))\int_{\partial B_1} D_i\Gamma(\bar{x}-y)\nu_j(y)dS_y \\
I_3 &= \int_{B_r(\hat{x})} D_{ij}\Gamma(x-y)(f(x)-f(y))dy \\
I_4 &= \int_{B_r(\hat{x})} D_{ij}\Gamma(\bar{x}-y)(f(y)-f(\bar{x}))dy \\
I_5 &= (f(x) - f(\bar{x}))\int_{B_1\setminus B_r(\hat{x})} D_{ij}\Gamma(x-y)dy \\
I_6 &= \int_{B_1\setminus B_r(\hat{x})}(D_{ij}\Gamma(x-y) - D_{ij}\Gamma(\bar{x}-y))(f(\bar{x})-f(y))dy
\end{aligned}$$

For I_1:

$$\begin{aligned}
|I_1| &\leq |f(x)| \cdot \int_{\partial B_1} |D_i\Gamma(x-y) - D_i\Gamma(\overline{x}-y)|dS_y \\
&\leq |f(x)| \cdot \int_{\partial B_1} |D(D_i\Gamma(\tilde{x}-y))| \cdot |x-\bar{x}|dS_y \\
&\leq |f(x)| \cdot |x-\bar{x}| \cdot \int_{\partial B_1} c|\tilde{x}-y|^{-n}dS_y \\
&\leq C|f(x)| \cdot |x-\bar{x}| \\
&\leq C|f(x)| \cdot |x-\bar{x}|^\alpha \cdot \left(\frac{1}{4}\right)^{1-\alpha}
\end{aligned}$$

where $\tilde{x} = \theta x + (1-\theta)\bar{x}$ and satisfies $|\tilde{x}-y| > \frac{1}{2}$ for any $y \in \partial B_1$.
For I_2:

$$\begin{aligned}
|I_2| &\leq |f(x) - f(\bar{x})| \int_{\partial B_1} |D_i\Gamma(\bar{x}-y)|dS_y \\
&\leq [f]_\alpha \cdot |x-\bar{x}|^\alpha \cdot \int_{\partial B_1} C|\bar{x}-y|^{1-n}dS_y \\
&\leq C\left(\frac{1}{2}\right)^{1-n} \cdot |\partial B_1| \cdot [f]_\alpha \cdot |x-\bar{x}|^\alpha
\end{aligned}$$

where $|\bar{x}-y| > \frac{1}{2}$ for any $y \in \partial B_1$.
For I_3 and I_4:

$$\begin{aligned}
|I_3| &\leq \int_{B_r(\hat{x})} |D_{ij}\Gamma(x-y)| \cdot |f(x)-f(y)| dy \\
&\leq \int_{B_{3/2r}(x)} C \cdot |x-y|^{-n} \cdot [f]_\alpha \cdot |x-y|^\alpha dy \\
&= C \cdot [f]_\alpha \cdot \int_0^{3/2r} t^{\alpha-n} \cdot \omega_n \cdot t^{n-1} dt \\
&= C \cdot [f]_\alpha \cdot \omega_n \cdot \frac{1}{\alpha} \cdot \left(\frac{3}{2}r\right)^\alpha \\
&= C \cdot \omega_n \cdot \frac{1}{\alpha} \cdot [f]_\alpha \cdot \left(\frac{3}{2}\right)^\alpha \cdot |x-\bar{x}|^\alpha
\end{aligned}$$

Similarly,

$$|I_4| \leq C \cdot \omega_n \cdot \frac{1}{\alpha} \cdot [f]_\alpha \cdot \left(\frac{3}{2}\right)^\alpha \cdot |x-\bar{x}|^\alpha$$

For I_5:

$$\begin{aligned}
|I_5| &= |f(x)-f(\bar{x})| \left| \int_{B_1 \setminus B_r(\hat{x})} D_{ij}\Gamma(x-y) dy \right| \\
&\leq [f]_\alpha \cdot |x-\bar{x}|^\alpha \cdot \left| \int_{\partial(B_1 \setminus B_r(\hat{x}))} D_i\Gamma(x-y) \cdot \nu_j(y) dS_y \right| \\
&\leq [f]_\alpha |x-\bar{x}|^\alpha \left(\int_{\partial B_1} |D_i\Gamma(x-y)| dS_y + \int_{\partial B_r(\hat{x})} |D_i\Gamma(x-y)| dS_y \right) \\
&\leq [f]_\alpha |x-\bar{x}|^\alpha \cdot \left(\int_{\partial B_1} C|x-y|^{1-n} dS_y + \int_{\partial B_r(\hat{x})} C|x-y|^{1-n} dS_y \right) \\
&\leq [f]_\alpha |x-\bar{x}|^\alpha \left(\int_{\partial B_1} C \cdot \left(\frac{1}{2}\right)^{1-n} dS_y + \int_{\partial B_r(\hat{x})} C \cdot \left(\frac{r}{2}\right)^{1-n} dS_y \right) \\
&\leq C \cdot [f]_\alpha \cdot |x-\bar{x}|^\alpha
\end{aligned}$$

For I_6:
Since

$$|D_{ij}\Gamma(x-y) - D_{ij}\Gamma(\bar{x}-y)| \leq |D(D_{ij}\Gamma(\tilde{x}-y))||x-\bar{x}| \leq C \cdot |\tilde{x}-y|^{-n-1}|x-\bar{x}|$$

where $\tilde{x} = \theta x + (1-\theta)\bar{x}$.Hence

$$\begin{aligned}
|I_6| &\leq \int_{B_1 \setminus B_r(\hat{x})} C \cdot |\tilde{x}-y|^{-n-1} \cdot |x-\bar{x}| \cdot [f]_\alpha \cdot |\bar{x}-y|^\alpha dy \\
&\leq C \cdot [f]_\alpha \cdot |x-\bar{x}| \cdot \int_{B_1 \setminus B_r(\hat{x})} \left(\frac{1}{2}|\hat{x}-y|\right)^{-n-1} \cdot (2|\hat{x}-y|)^\alpha dy \\
&\leq C \cdot 2^{n+1+\alpha} \cdot [f]_\alpha \cdot |x-\bar{x}| \cdot \int_{|y-\hat{x}| \geq r} |\hat{x}-y|^{-n-1+\alpha} dy \\
&= C \cdot 2^{n+1+\alpha} \cdot [f]_\alpha \cdot |x-\bar{x}| \cdot \int_r^\infty t^{-n-1+\alpha} \cdot \omega_n \cdot t^{n-1} dt \\
&= C \cdot \omega_n \cdot 2^{n+1+\alpha} \cdot \frac{1}{1-\alpha} \cdot [f]_\alpha \cdot |x-\bar{x}|^\alpha
\end{aligned}$$

where we have the fact that

$$|\bar{x}-y| \leq |\tilde{x}-y| + |\bar{x}-\hat{x}| \leq |\hat{x}-y| + |\hat{x}-y|$$

$$|\hat{x}-y| \leq |\tilde{x}-y| + |\hat{x}-\tilde{x}| \leq |\tilde{x}-y| + |\tilde{x}-y|$$

Thus we have

$$|D_{ij}w(x) - D_{ij}w(\bar{x})| \leq C(|f(x)| + [f]_\alpha)|x-\bar{x}|^\alpha$$

This finishes the proof. □

Remark 1.3 *Theorem holds with B_1 and $B_{1/2}$ replaced by any bounded C^1-domain Ω and any $\Omega' \subset\subset \Omega$,respectively.*

Theorem 1.3 *Let $f \in C^\alpha(B_1)$,for some $\alpha \in (0,1)$,with $|f|_{C^\alpha(B_1)} < \infty$.Suppose $u \in L^\infty(B_1) \cap C^2(B_1)$ satisfies $\Delta u = f$ in B_1.Then $u \in C^{2,\alpha}(B_1)$.Moreover*

$$|u|_{C^{2,\alpha}(B_{1/2})} \leq C(|u|_{L^\infty(B_1)} + |f|_{C^\alpha(B_1)})$$

where C is a positive constant depending only on n and α.

Proof Let w be the Newtonian potential of f in B_1.By **Lemma1.2**,we have

$$|w|_{C^1(B_1)} \leq C|f|_{L^\infty(B_1)}$$

By **Lemma1.3**,$w \in C^{2,\alpha}(B_1)$,$\Delta w = f$ in B_1 and

$$|w|_{C^{2,\alpha}(B_{1/2})} \leq C|f|_{C^\alpha(B_1)}$$

Set $v = u - w$.Then v is a harmonic function in B_1 and

$$|v|_{L^\infty(B_1)} \leq |u|_{L^\infty(B_1)} + |w|_{L^\infty(B_1)} \leq |u|_{L^\infty(B_1)} + C|f|_{L^\infty(B_1)}$$

Interior gradient estimates?? on derivatives of harmonic functions imply

$$|v|_{C^{2,\alpha}(B_{1/2})} \leq C|v|_{L^\infty(B_1)} \leq C(|u|_{L^\infty(B_1)} + |f|_{L^\infty(B_1)})$$

Hence,we obtain

$$\begin{aligned}
|u|_{C^{2,\alpha}(B_{1/2})} &\leq |v|_{C^{2,\alpha}(B_{1/2})} + |w|_{C^{2,\alpha}(B_{1/2})} \\
&\leq C(|u|_{L^\infty(B_1)} + |f|_{L^\infty(B_1)}) + C|f|_{C^\alpha(B_1)} \\
&\leq C(|u|_{L^\infty(B_1)} + |f|_{C^\alpha(B_1)})
\end{aligned}$$

This finishes the proof. □

Theorem 1.4 *Let $B_R(x_0)$ be a ball in $\mathbb{R}^n$ and $f \in C^\alpha(B_R(x_0))$,for some $\alpha \in (0,1)$,with $|f|_{C^\alpha(B_R(x_0))} < \infty$.Suppose $u \in L^\infty(B_R(x_0)) \cap C^2(B_R(x_0))$ satisfies $\Delta u = f$ in $B_R(x_0)$.Then $u \in C^{2,\alpha}(B_R(x_0))$.Moreover,*

$$\begin{aligned}
&R|Du|_{L^\infty(B_{R/2}(x_0))} + R^2|D^2u|_{L^\infty(B_{R/2}(x_0))} + R^{2+\alpha}[D^2u]_{C^\alpha(B_{R/2(x_0)})} \\
\leq &C(|u|_{L^\infty(B_R)(x_0)} + R^2|f|_{L^\infty(B_R)(x_0)} + R^{2+\alpha}[f]_{C^\alpha(B_R)(x_0)})
\end{aligned}$$

where C is a positive constant depending only on n and α.

Proof Set $\tilde{u}(x) = u(Rx+x_0)$ and $\tilde{f}(x) = R^2 f(Rx+x_0)$,then $\tilde{u}(x) \in L^\infty(B_1) \cap C^2(B_1)$ and $\tilde{f}(x) \in C^\alpha(B_1)$. **Theorem1.2** implies that $\tilde{u}(x) \in C^{2,\alpha}(B_1)$,hence $u \in C^{2,\alpha}(B_R(x_0))$.Moreover

$$|\tilde{u}|_{C^{2,\alpha}(B_{1/2})} \leq C(|\tilde{u}|_{L^\infty(B_1)} + |\tilde{f}|_{C^\alpha(B_1)})$$

Then

$$|u(Rx+x_0)|_{C^{2,\alpha}(B_{1/2})} \leq C\left(|u(Rx+x_0)|_{L^\infty(B_1)} + R^2|f(Rx+x_0)|_{C^\alpha(B_1)}\right)$$

$\Longrightarrow$

$$\begin{aligned}
&\sup_{B_{1/2}}|u(Rx+x_0)| + \sup_{B_{1/2}}|Du(Rx+x_0)| + \sup_{B_{1/2}}|D^2u(Rx+x_0)| + \sup_{B_{1/2}}\frac{|D^2u(Rx+x_0)-D^2u(Ry+x_0)|}{|R(x-y)|^\alpha}\\
\leq& C\left(|u(Rx+x_0)|_{L^\infty(B_1)} + R^2\left(\sup_{B_1}|f(Rx+x_0)| + \sup_{B_1}\frac{|f(Rx+x_0)-f(Ry+x_0)|}{|R(x-y)|^\alpha}\right)\right)
\end{aligned}$$

$\Longrightarrow$

$$\begin{aligned}
&\sup_{B_{R/2}(x_0)}|u| + R\sup_{B_{R/2}(x_0)}|Du| + R^2\sup_{B_{R/2}(x_0)}|D^2u| + \sup_{B_{R/2}(x_0)}\frac{R^2|D^2u(x)-D^2u(y)|}{R^\alpha|x-y|^\alpha}\\
\leq& C\left(|u|_{L^\infty(B_R(x_0))} + R^2\sup_{B_R(x_0)}|f| + R^2\sup_{B_R(x_0)}\frac{|f(x)-f(y)|}{R^\alpha|x-y|^\alpha}\right)
\end{aligned}$$

We obtain

$$\begin{aligned}
&R|Du|_{L^\infty(B_{R/2}(x_0))} + R^2|D^2u|_{L^\infty(B_{R/2}(x_0))} + R^{2+\alpha}[D^2u]_{C^\alpha(B_{R/2(x_0)})}\\
\leq& C(|u|_{L^\infty(B_R)(x_0)} + R^2|f|_{L^\infty(B_R)(x_0)} + R^{2+\alpha}[f]_{C^\alpha(B_R)(x_0)})
\end{aligned}$$

□

Remark 1.4 *We point out that the estimate in* ***Theorem1.4*** *holds when* $\frac{R}{2}$ *in the left hand side is replaced by any* $r \in (0,R)$.

Corollary 1.1 *Let* B *be a ball in* $\mathbb{R}^n$ *and* $f \in C^\alpha(B)$,*for some* $\alpha \in (0,1)$,*with* $|f|_{C^\alpha(B)} < \infty$.*Suppose* $u_k \in L^\infty(B) \cap C^2(B)$ *satisfies* $\Delta u_k = f$ *in* B *for* $k = 1,2,\cdots$,*and*

$$\sup_{k\geq 1}|u_k|_{L^\infty(B)} < \infty$$

Then there exist a subsequence $u_{k'}$ *and a* $u \in C^{2,\alpha}(B)$ *such that* $u_{k'} \longrightarrow u$ *in* C^2 *in any compact subset of* B *and* $\Delta u = f$ *in* B.

Proof Set $M = \sup_{k\geq 1}|u_k|_{L^\infty(B)}$.By **Theorem1.4** and the remark,$u_k \in C^{2,\alpha}(B)$ and,for any $B' \subset\subset B$,

$$|u_k|_{C^{2,\alpha}(B')} \leq C(M + |f|_{C^\alpha(B)})$$

Hence,the second derivatives of $\{u_k\}$ are equicontinuous on compact subsets of B.We obtain the desired result by **Arzela Theorem** and a diagonalization process. □

1.4 Dirichlet problem

In this section,we solve the Dirichlet problem for the Poisson equation in balls.

Theorem 1.5 *Let B be a ball in $\mathbb{R}^n$.Then,for any $f \in C^\alpha(B)$,for some $\alpha \in (0,1)$,with $|f|_{C^\alpha(B)} < \infty$,and any $\varphi \in C(\partial B)$,there exists a (unique) solution $u \in C(\bar{B}) \cap C^{2,\alpha}(B)$ of the Dirichlet problem*

$$\begin{cases} \Delta u = f & in \quad B \\ u = \varphi & on \quad \partial B \end{cases}$$

Proof Let w be the Newtonian potential of f in B.By **Lemma1.3**,$w \in C(\bar{B}) \cap C^{2,\alpha}(B)$ and $\Delta w = f$ in B.Now consider

$$\begin{cases} \Delta v = 0 & in \quad B \\ v = \varphi - w & on \quad \partial B \end{cases}$$

Then exist a solution $v \in C(\bar{B}) \cap C^\infty(B)$.Then $u = v + w$ is the desired solution of the Dirichlet problem. □
Now we allow f to be unbounded.For any ball B in $\mathbb{R}^n$,we write

$$d_x = \mathrm{dist}(x, \partial B)$$

Theorem 1.6 *Let B be a ball in $\mathbb{R}^n$ and $f \in C^\alpha(B)$,for some $\alpha \in (0,1)$,satisfy*

$$\sup_{x \in B} d_x^{2-\beta} |f(x)| < \infty$$

for some $\beta \in (0,1)$.Then there exists a solution $u \in C(\bar{B}) \cap C^{2,\alpha}(B)$ of the Dirichlet problem

$$\begin{cases} \Delta u = f & in \quad B \\ u = 0 & on \quad \partial B \end{cases}$$

Proof For any integer m,set

$$f_m = \begin{cases} m & if \quad f \geq m \\ f & if \quad |f| \leq m \\ -m & if \quad f \leq -m \end{cases}$$

Then,$f_m \in C^\alpha(B)$ and $|f_m|_{C^\alpha(B)} < \infty$.Let $u_m \in C(\bar{B}) \cap C^{2,\alpha}(B)$ be the solution of

$$\begin{cases} \Delta u_m = f_m & in \quad B \\ u_m = 0 & on \quad \partial B \end{cases}$$

The existence of u_m is guaranteed by **Theorem1.5**.By **Lemma1.2**,we have

$$\sup_{x \in B} d_x^{-\beta} |u_m(x)| \leq C \sup_{x \in B} d_x^{2-\beta} |f_m(x)| \leq C \sup_{x \in B} d_x^{2-\beta} |f(x)|$$

Set $B = B_R$ and consider an increasing sequence $r_k \longrightarrow R$ such that $|f| \leq k$ in B_{r_k}.
Hence,$\{u_m\}$ is uniformly bounded and $\Delta u_m = f$ in B_{r_k},for $k \leq m$.
In B_{r_k},a uniform interior $C^{2,\alpha}$-estimate holds for $\{u_m\}$(**Theorem1.3**).By **Corollary1.1**,a subsequence of $\{u_m\}$ is convergent in C^2 in any compact subset of B_{r_k}.
By applying this successively to $\{B_{r_k}\}$,a subsequence of $\{u_m\}$ converges to a $u \in C^{2,\alpha}(B)$ in C^2-norm.Then,$\Delta u = f$ in B and

$$d_x^{-\beta} |u(x)| \leq C \sup_{x \in B} d_x^{2-\beta} |f(x)|$$

Hence $u = 0$ on ∂B. $\square$

We now discuss regularity of solutions up to the boundary.

Lemma 1.4 *Let $\Omega = B_1^+$ in $\mathbb{R}_+^n$ and $f \in L^\infty(B_1^+) \cap C^\alpha(B_1^+ \cup \Sigma_1)$ for some $\alpha \in (0,1)$.Suppose w is the Newtonian potential of f in B_1^+.Then $w \in C(\bar{B}_1^+) \cap C^{2,\alpha}(B_1^+ \cup \Sigma_1)$ and $\Delta w = f$ in B_1^+.Moreover,*

$$|w|_{C^{2,\alpha}(B_{1/2}^+ \cup \Sigma_{1/2})} \leq C|f|_{C^\alpha(B_1^+ \cup \Sigma_1)}$$

where C is a positive constant depending only on n and α.

Proof It is trivial by **Lemma1.3**. $\square$

We now discuss arbitrary solutions of the Poisson equation.

Theorem 1.7 *Let $f \in L^\infty(B_1^+) \cap C^\alpha(B_1^+ \cup \Sigma_1)$ for some $\alpha \in (0,1)$.Suppose $u \in L^\infty(B_1^+) \cap C^2(B_1^+)$ satisfies*

$$\begin{cases} \Delta u = f & in \quad B_1^+ \\ u = 0 & on \quad \Sigma_1 \end{cases}$$

Then $u \in C^{2,\alpha}(B_1^+ \cup \Sigma_1)$.Moreover

$$|u|_{C^{2,\alpha}(B_{1/2}^+ \cup \Sigma_1)} \leq C(|u|_{L^\infty(B_1^+)} + |f|_{C^\alpha(B_1^+ \cup \Sigma_1)})$$

where C is a positive constant depending only on n and α.

Proof For $x = (x', x_n)$,set $x^* = (x', -x_n)$ and

$$B_1^- = B_1 \cap \{x_n < 0\}$$

Set

$$f^*(x) = f^*(x', x_n) = \begin{cases} f(x', x_n) & x_n \geq 0 \\ f(x', x_n) & x_n < 0 \end{cases}$$

Then $f^* \in C^\alpha(B_1)$ and

$$|f^*|_{C^\alpha(B_1)} \leq 2|f|_{C^\alpha(B_1^+ \cup \Sigma_1)}$$

Let w be the Newtonian potential of f in B_1^+,i.e,

$$w(x) = \int_{B_1^+} \Gamma(x-y) f(y) dy$$

By **Lemma1.2**,

$$|w|_{C^1(B_1^+)} \leq C|f|_{L^\infty(B_1^+)}$$

By **Lemma1.4**,$w \in C^{2,\alpha}(B_1^+ \cup \Sigma_1)$ and $\Delta w = f$ in B_1^+.Moreover,

$$|w|_{C^{2,\alpha}(B_{1/2}^+)} \leq C|f|_{C^\alpha(B_1^+ \cup \Sigma_1)}$$

Let w^* be the Newtonian potential of f^* in B_1,i.e,

$$w^*(x) = \int_{B_1} \Gamma(x-y) f^*(y) dy$$

Then,

$$|w^*|_{C^1(B_1)} \leq C|f^*|_{L^\infty(B_1)} \leq C|f|_{L^\infty(B_1^+)}$$

By **Lemma1.2**,$w^* \in C^{2,\alpha}(B_1)$,$\Delta w^* = f^*$ in B_1.Moreover,

$$|w^*|_{C^{2,\alpha}(B_{1/2})} \leq C|f^*|_{C^\alpha(B_1)} \leq C|f|_{C^\alpha(B_1^+\cup\Sigma_1)}$$

Set $\tilde{w} = 2w - w^*$.Then

$$\begin{aligned}\tilde{w}(x) &= 2\int_{B_1^+}\Gamma(x-y)f(y)dy - \int_{B_1}\Gamma(x-y)f^*(y)dy \\ &= \int_{B_1^+}\Gamma(x-y)f(y)dy - \int_{B_1^-}\Gamma(x-y)f^*(y)dt\end{aligned}$$

By

$$\int_{B_1^-}\Gamma(x-y)f^*(y)dy = \int_{B_1^+}\Gamma(x-y^*)f(y)dy$$

we obtain

$$\tilde{w}(x) = \int_{B_1^+}(\Gamma(x-y) - \Gamma(x-y^*))f(y)dy$$

Then $\tilde{w}(x) = 0$ on Σ_1.
In conclusion,

$$\begin{cases}\Delta\tilde{w} = f & in \quad B_1^+ \\ \tilde{w}(x) = 0 & on \quad \Sigma_1\end{cases}$$

and

$$|v|_{L^\infty(B_1^+)} \leq |u|_{L^\infty(B_1^+)} + |w|_{L^\infty(B_1^+)} \leq |u|_{L^\infty(B_1^+)} + C|f|_{L^\infty(B_1^+)}$$

By reflection,v can be extended to a harmonic function in B_1 and the **interior estimates** on derivatives of harmonic functions imply

$$|v|_{C^{2,\alpha}(B_{1/2})} \leq C|v|_{L^\infty(B_1^+)} \leq C(|u|_{L^\infty(B_1^+)} + |f|_{L^\infty(B_1^+)})$$

Then,we have

$$|u|_{C^{2,\alpha}(B_{1/2}^+\cup\Sigma_{1/2})} \leq C(|u|_{L^\infty(B_1^+)} + |f|_{C^\alpha(B_1^+\cup\Sigma_1)})$$

The desired estimate for u follows easily from a simple covering. □

We solve the Dirichlet problem for the Poisson equation in balls if boundary values are $C^{2,\alpha}$.

Theorem 1.8 *Let B be a ball in $\mathbb{R}^n$.Then for any $f \in C^\alpha(\bar{B})$ and $\varphi \in C^{2,\alpha}(\partial B)$,for some $\alpha \in (0,1)$,there exists a (unique) solution $u \in C^{2,\alpha}(\bar{B})$ of the Dirichlet problem*

$$\begin{cases}\Delta u = f & in \quad B \\ u = \varphi & on \quad \partial B\end{cases}$$

Proof By **Theorem1.5**,there exists a solution $u \in C(\bar{B}) \cap C^{2,\alpha}(B)$.We need to prove $u \in C^{2,\alpha}(\bar{B})$.
Without loss of generality,we assume $\varphi = 0$.Otherwise,we consider $\tilde{u} = u - \varphi$.Then $\Delta\tilde{u} = f - \Delta\varphi$ in B and $\tilde{u} = 0$ on ∂B.

By a translation and a rotation,we assume B is given by

$$|x'|^2 + (x_n - R)^2 = R^2$$

In other words,B has center at $(0, \cdots, 0, R)$ and radius R.In particular,the origin is on ∂B.Consider the transformation

$$y = \frac{x}{|x|^2}$$

Under this transformation,B is transformed onto $\Omega = \{y_n > \frac{1}{2R}\}$ and $\partial B \setminus \{0\}$ is transformed onto $\partial\Omega = \{y_n = \frac{1}{2R}\}$.Set

$$v(y) = |x|^{n-2}u(x)$$

Then $v \in C(\bar{\Omega}) \cap C^2(\Omega)$ and $v = 0$ on $\partial\Omega$,since $u \in C(\bar{B}) \cap C^2(B)$ and $u = 0$ on ∂B.A straightforward calculation yields

$$\Delta_y v(y) = |y|^{-n-2}\Delta_x u(x) = |y|^{-n-2} f\left(\frac{y}{|y|^2}\right)$$

By **Theorem1.7**,$v \in C^{2,\alpha}(\bar{\Omega})$.Since any point of ∂B may be taken as the origin,we have $u \in C^{2,\alpha}(\bar{B})$. □

2 Schauder Theory—Non-divergence Form

In this chapter,we discuss the Schauder theory for uniformly elliptic linear equations of non-divergence form.Three main topics are a priori estimates in Hölder norms,regularity of arbitrary solutions and the solvability of the Dirichlet problem.Among these topics,a priori estimates are the most fundamental and the basis for the existence and regularity of solutions.We will discuss both the interior Schauder theory and the global Schauder theory.

In this chapter,we let Ω be a domain in $\mathbb{R}^n$,bounded most of the time,and let a_{ij},b_i and c be defined in Ω,with $a_{ij} = a_{ji}$.We consider the operator L given by

$$Lu = a_{ij}D_{ij}u + b_iD_iu + cu \quad \text{in } \Omega,$$

for any $u \in C^2(\Omega)$.The operator L is always assumed to be **strictly elliptic** in Ω,namely,

$$a_{ij}\xi_i\xi_j \geq \lambda|\xi|^2 \quad \forall x \in \Omega, \xi \in \mathbb{R}^n,$$

for some positive constant λ.

In this section,we discuss the interior Schauder theory.Three main topics are a interior priori estimates in Hölder norms,interior regularity of arbitrary solutions and the solvability of the Dirichlet problem for continuous boundary values.

2.1 Interior $C^{2,\alpha}$ Schauder estimates

We will derive interior Schauder estimates,which yield estimates of $C^{2,\alpha}$-norms of solutions in compact subsets,in terms of the L^∞-norms of solutions and the C^α-norms of the nonhomogeneous terms,if coefficients are also C^α,for some $\alpha \in (0, 1)$.

Giving some notations.

Let B_R be a ball in $\mathbb{R}^n$,k be a nonnegative integer and $\alpha \in (0,1)$.We define

$$|u|^*_{C^{k,\alpha}(B_R)} = \sum_{i=0}^{k} R^i |D^i u|_{L^\infty(B_R)} + R^{k+\alpha}[D^k u]_{C^\alpha(B_R)}$$

We note that factors of R results from scaling.In particular,

$$\begin{aligned}
|u|^*_{C^{\alpha}(B_R)} &= |u|_{L^\infty} + R^\alpha [u]_{C^\alpha}, \\
|u|^*_{C^{1,\alpha}(B_R)} &= |u|_{L^\infty} + R \cdot |Du|_{L^\infty} + R^{1+\alpha}[Du]_{C^\alpha}, \\
|u|^*_{C^{2,\alpha}(B_R)} &= |u|_{L^\infty} + R \cdot |Du|_{L^\infty} + R^2 \cdot |D^2 u|_{L^\infty} + R^{2+\alpha}[D^2 u]_{C^\alpha}.
\end{aligned}$$

Proving an interpolation inequality for Hölder continuous functions.

Lemma 2.1 (Interpolation Inequality) *Let $\alpha, \mu \in (0,1)$ and B_R be a ball of radius R in $\mathbb{R}^n$.Then,*
(1)for any $u \in C^1(\overline{B_R})$,

$$\mu^\alpha R^\alpha [u]_{C^\alpha(B_R)} \le C(\mu R |Du|_{L^\infty(B_R)} + |u|_{L^\infty(B_R)}),$$

(2)for any $u \in C^{1,\alpha}(\overline{B_R})$,

$$\mu R |Du|_{L^\infty(B_R)} \le C(\mu^{1+\alpha} R^{1+\alpha}[Du]_{C^\alpha(B_R)} + |u|_{L^\infty(B_R)}),$$

(3)for any $u \in C^2(\overline{B_R})$,

$$\mu R |Du|_{L^\infty(B_R)} \le C(\mu^2 R^2 |D^2 u|_{L^\infty(B_R)} + |u|_{L^\infty(B_R)}),$$

(4)for any $u \in C^{2,\alpha}(\overline{B_R})$,

$$\sum_{i=1}^{2} (\mu R)^i |D^i u|_{L^\infty(B_R)} + \sum_{i=0}^{1} (\mu R)^{i+\alpha}[D^i u]_{C^\alpha(B_R)} \le C((\mu R)^{2+\alpha}[D^2 u]_{C^\alpha(B_R)} + |u|_{L^\infty(B_R)}).$$

Proof Set $v(x) = u(Rx)$,so we will consider $R = 1$ only.
(1)For any $x, y \in B_1$,since $u \in C^1(\overline{B_R})$,then exists $\xi \in B_R$,such that

$$\frac{|u(x) - u(y)|}{|x-y|^\alpha} = |Du(\xi)|.$$

Then,we have

$$\frac{|u(x) - u(y)|}{|x-y|^\alpha} \le |Du|_{L^\infty(B_1)} |x-y|^{1-\alpha}.$$

If $|x-y| \le \mu$,then

$$\frac{|u(x) - u(y)|}{|x-y|^\alpha} \le \mu^{1-\alpha} |Du|_{L^\infty(B_1)}.$$

If $|x-y| \ge \mu$,then

$$\frac{|u(x) - u(y)|}{|x-y|^\alpha} \le 2\frac{|u|_{L^\infty(B_1)}}{\mu^\alpha} = 2\mu^{-\alpha}|u|_{L^\infty(B_1)}.$$

Hence,

$$[u]_{C^\alpha(B_1)} \le \mu^{1-\alpha}|Du|_{L^\infty(B_1)} + 2\mu^{-\alpha}|u|_{L^\infty(B_1)}.$$

Then,

$$\mu^\alpha [u]_{C^\alpha(B_1)} \le \mu |Du|_{L^\infty(B_1)} + 2|u|_{L^\infty}.$$

(2)For any $x \in B_1$,set $\mathbf{e} = -\frac{\mathbf{x}}{|\mathbf{x}|}(|\mathbf{x}| \neq 0)$,then $x + \mu\eta \in B_1$ for any unit vector $\boldsymbol{\eta} \in S^{n-1}$ with $\text{Angle}(\boldsymbol{\eta}, \mathbf{e}) < \frac{\pi}{4}$ and any $\mu \in (0, 1)$.
Now fix such $bm\eta$ and μ,there exists an $\bar{x}$ between x and $x + \mu\eta$ such that

$$|D_{\boldsymbol{\eta}} u(\bar{x})| = \frac{|u(x) - u(x + \mu\eta)|}{\mu} \leq \frac{2}{\mu}|u|_{L^\infty(B_1)}.$$

Hence,

$$|D_{\boldsymbol{\eta}} u(x)| = |D_{\boldsymbol{\eta}} u(\bar{x}) + D_{\boldsymbol{\eta}} u(x) - D_{\boldsymbol{\eta}} u(\bar{x})| \leq \frac{2}{\mu}|u|_{L^\infty(B_1)} + [Du]_{C^\alpha(B_1)}|x - \bar{x}|^\alpha.$$

Note $|x - \bar{x}| \leq \mu$,then

$$|D_{\boldsymbol{\eta}} u(x)| \leq \frac{2}{\mu}|u|_{L^\infty(B_1)} + \mu^\alpha [Du]_{C^\alpha(B_1)}.$$

If $|Du(x)| \neq 0$,set $\boldsymbol{\eta_0} = \frac{Du(x)}{|Du(x)|}$,then there exists $\boldsymbol{\eta} \in S^{n-1}$,$\text{Angle}(\boldsymbol{\eta}, \mathbf{e}) < \frac{\pi}{4}$,such that $\text{Angle}(\boldsymbol{\eta}, \boldsymbol{\eta_0}) \leq \frac{\pi}{4}$ or $\text{Angle}(\boldsymbol{\eta}, -\boldsymbol{\eta_0}) \leq \frac{\pi}{4}$**(Why?)**.Thus,we obtain

$$\begin{aligned}
|D_{\boldsymbol{\eta}} u(x) = |Du(x) \cdot \boldsymbol{\eta}| &= \left| |Du(x)| \cdot \frac{Du(x)}{|Du(x)|} \cdot \boldsymbol{\eta} \right| \\
&= |Du(x)||\boldsymbol{\eta_0} \cdot \boldsymbol{\eta}| = |Du(x)||\boldsymbol{\eta_0}||\boldsymbol{\eta}| \cdot \cos \text{Angle}(\boldsymbol{\eta_0}, \boldsymbol{\eta}) \\
&\geq \frac{\sqrt{2}}{2} \geq |Du(x)|,
\end{aligned}$$

$\Longrightarrow$

$$|Du(x)| \leq \frac{|D_{\boldsymbol{\eta} u(x)}|}{\sqrt{2}/2} \leq \frac{2\sqrt{2}}{\mu}|u(x)|_{L^\infty(B_1)} + \sqrt{2}\mu^\alpha [Du]_{C^\alpha(B_1)}.$$

Then

$$\mu|Du(x)|_{L^\infty(B_1)} \leq 2\sqrt{2}|u(x)|_{L^\infty(B_1)} + \sqrt{2}\mu^{\alpha+1}[Du]_{C^\alpha(B_1)}.$$

(3)The proof is a simple modification of that of (2).
(4)From (1),(2) and (3),we will only prove that

$$\mu^2|D^2 u|_{L^\infty(B_1)} \leq C(\mu^{2+\alpha}[D^2 u]_{C^\alpha(B_1)} + |u|_{L^\infty(B_1)}).$$

In fact,

$$\begin{aligned}
\mu^\alpha [u]_{C^\alpha(B_1)} &\leq C_1(\mu|Du|_{L^\infty(B_1)} + |u|_{L^\infty(B_1)}) \\
&\leq C_1(C_2(\mu^2|D^2 u|_{L^\infty(B_1)} + |u|_{L^\infty(B_1)}) + |u|_{L^\infty(B_1)}) \\
&\leq C_3(\mu^2|D^2 u|_{L^\infty(B_1)} + |u|_{L^\infty(B_1)}) \\
&\leq C(\mu^{2+\alpha}[D^2 u]_{C^\alpha(B_1)} + |u|_{L^\infty(B_1)}),
\end{aligned}$$

$$\begin{aligned}
\mu|Du|_{L^\infty(B_1)} &\leq C_1(\mu^2|D^2 u|_{L^\infty(B_1)} + |u|_{L^\infty(B_1)}) \\
&\leq C(\mu^{2+\alpha}[D^2 u]_{C^\alpha(B_1)} + |u|_{L^\infty(B_1)}),
\end{aligned}$$

and

$$\begin{aligned}
\mu^{1+\alpha}[Du]_{C^\alpha(B_1)} &\leq \mu C_1(\mu|D^2 u|_{L^\infty(B_1)} + |Du|_{L^\infty(B_1)}) \\
&= C_1(\mu^2|D^2 u|_{L^\infty(B_1)} + \mu|Du|_{L^\infty(B_1)})
\end{aligned}$$

$$\leq C(\mu^{2+\alpha}[D^2u]_{C^\alpha(B_1)} + |u|_{L^\infty(B_1)}).$$

Then,we have the desired result.
By applying (2),we have

$$\mu|D^2u|_{L^\infty(B_1)} \leq C_1(|Du|_{L^\infty(B_1)} + \mu^{1+\alpha}[D^2u]_{C^\alpha(B_1)}).$$

By applying (3),we have

$$\tau|Du|_{L^\infty(B_1)} \leq C_2(|u|_{L^\infty(B_1)} + \tau^2|D^2u|_{L^\infty(B_1)}).$$

Then

$$\mu^2|D^2u|_{L^\infty(B_1)} \leq C_1\left(\frac{C_2\mu}{\tau}|u|_{L^\infty(B_1)} + \mu C_2\cdot\tau|D^2u|_{L^\infty(B_1)} + \mu^{2+\alpha}[D^2u]_{C^\alpha(B_1)}\right).$$

Choose $C_1C_2\mu\tau = \frac{1}{2}\mu^2$,then $\tau = \frac{\mu}{2C_1C_2}$.Hence

$$\mu^2|D^2u|_{L^\infty(B_1)} \leq C(|u|_{L^\infty(B_1)} + \mu^{2+\alpha}[D^2u]_{C^\alpha(B_1)}).$$

□

Now we can give the interior $C^{2,\alpha}$ Schauder estimates,take some special cases first,and then give the general.

• Elliptic equations with constant leading coefficients.

Lemma 2.2 *Let $f \in C^\alpha(B_R)$,for some $\alpha \in (0,1)$,and (a_{ij}) be a constant symmetric $n\times n$ matrix satisfying*

$$\lambda|\xi|^2 \leq a_{ij}\xi_i\xi_j \leq \Lambda|\xi|^2 \quad \forall\xi\in\mathbb{R}^n,$$

for some positive constants λ and Λ.Suppose $u \in C^2(B_R)$ satisfies

$$a_{ij}D_{ij}u = f \quad \text{in } B_R.$$

Then,$u \in C^{2,\alpha}(B_R)$.Moreover,

$$|u|^*_{C^{2,\alpha}(B_{R/2})} \leq C\left(|u|_{L^\infty(B_R)} + R^2|f|^*_{C^\alpha(B_R)}\right),$$

where C is a positive constant depending only on n,α,λ and Λ.

Proof The desired estimate for the case $a_{ij} = \delta_{ij}$ follows from **Theorem1.4**.
Now we consider the general case.Set $A = (a_{ij})$.Then there exists an orthogonal matrix P such that

$$P'AP = \text{diag}\{\lambda_1,\lambda_2,\cdots,\lambda_n\},$$

with $\lambda \leq \lambda_i \leq \Lambda$ for $i = 1,2,\cdots,n$.
Now set $Q = PD$ with

$$D = \text{diag}\{\lambda_1^{-1/2},\lambda_2^{-1/2},\cdots,\lambda_n^{-1/n}\},$$

then $Q'AQ = I_n$.
Consider the transform $y = Qx$,then $x = Q^{-1}y$ and set $\tilde{u}(y) = u(x)$,$\tilde{f}(y) = f(x)$,for $y \in Q(B_R)$.Then

$$\tilde{u}(y) = u(x) = u(Q^{-1}y), \quad \tilde{f}(y) = f(x) = f(Q^{-1}y),$$

and $\tilde{u}$ satisfies that

$$\begin{aligned}\Delta\tilde{u}(y) &= \sum_{k=1}^{n}\frac{\partial^2\tilde{u}(y)}{\partial y_k^2} = \sum_{k=1}^{n}\left(\sum_{i,j=1}^{n}\frac{\partial^2 u}{\partial x_i\partial x_j}\cdot\frac{\partial x_i}{\partial y_k}\cdot\frac{\partial x_i}{\partial y_k}\right)\\ &= \sum_{i,j=1}^{n}\left(\frac{\partial^2 u}{\partial x_i\partial x_j}\cdot\left(\sum_{k=1}^{n}\frac{\partial x_i}{\partial y_k}\cdot\frac{\partial x_j}{\partial y_k}\right)\right)\\ &= \mathrm{tr}(D^2u\cdot(Q^{-1})'\cdot(Q^{-1}))\\ &= \mathrm{tr}(A\cdot D^2u) = \sum_{i,j=1}^{n}a_{ij}u_{ij} = f(x) = \tilde{f}(y)\end{aligned}$$

for any $y\in Q(B_R)$.
Since the orthogonal matrix P preserves length,we have

$$|Qx| = |PDx| = |Dx| = \left(\sum_{i=1}^{n}\lambda_i^{-1}x_i^2\right)^{1/2},$$

then

$$\Lambda^{-1/2}|x| \le |Qx| \le \lambda^{-1/2}|x|.$$

It is obvious that

$$\mathrm{dist}(Q(B_{R/2}),\partial Q(B_R)) \ge \Lambda^{-1/2}\frac{R}{2}.$$

In fact,$\forall x\in B_{R/2}$ and $x'\in\partial B_R$,we have

$$|Q(x-x')| \ge \Lambda^{-1/2}|x-x'| \ge \Lambda^{-1/2}\frac{R}{2}.$$

Choose constant $\tau = \min\left(\frac{1}{2}\Lambda^{-1/2},\frac{1}{2}\right)\in(0,1)$.Since

$$Q(\overline{B_{R/2}}) \subset \bigcup_{y_i\in Q(B_{R/2})} B_{\tau R/2}(y_i),$$

then exists a positive integer N,depending only on n,λ and Λ,such that

$$Q(B_{R/2}) \subset \bigcup_{i=1}^{N} B_{\tau R/2}(y_i) \subset \bigcup_{i=1}^{N} B_{\tau R}(y_i).$$

Since $\mathrm{dist}(y_i,\partial Q(B_R)) \ge \Lambda^{-1/2}\frac{R}{2} = \tau R$,then $B_{\tau R}(y_i)\subset Q(B_R)$,$\forall i = 1,2,\cdots,N$.Thus

$$Q(B_{R/2}) \subset \bigcup_{i=1}^{N} B_{\tau R/2}(y_i) \subset \bigcup_{i=1}^{N} B_{\tau R}(y_i) \subset Q(B_R).$$

Apply **Theorem1.4** for $\tilde{u}$ in $B_{\tau R}(y_i)$,then

$$|\tilde{u}|^*_{C^{2,\alpha}(B_{\tau R/2}(y_i))} \le C\left(|\tilde{u}|_{L^\infty(B_{\tau R}(y_i))} + R^2|\tilde{f}|^*_{C^\alpha(B_{\tau R}(y_i))}\right).$$

We have

$$\begin{aligned}|\tilde{u}|^*_{C^{2,\alpha}(Q(B_{R/2}))} &\le |\tilde{u}|^*_{C^{2,\alpha}(\cup_{i=1}^N B_{\tau R/2}(y_i))}\\ &\le \sum_{i=1}^N |\tilde{u}|^*_{C^{2,\alpha}(B_{\tau R/2}(y_i))}\\ &\le C\cdot N(|\tilde{u}|_{L^\infty(Q(B_R))} + R^2|\tilde{f}|^*_{C^\alpha(Q(B_R))}).\end{aligned}$$

We then have the desired result that

$$|u|^*_{C^{2,\alpha}(B_{R/2})} \le C(n,\lambda,\Lambda)\left(|u|_{L^\infty(B_R)} + R^2|f|^*_{C^\alpha(B_R)}\right).$$

□

• **Uniformly elliptic equations—Freeze coefficients.**

Lemma 2.3 *Let $\Omega = B_R$ be a ball in $\mathbb{R}^n$ and L be an elliptic operator in B_R.Assume a_{ij}, b_i and c are C^α-functions in B_R,for some $\alpha \in (0,1)$,and satisfy*

$$a_{ij}\xi_i\xi_j \ge \lambda|\xi|^2 \quad \forall x \in \Omega, \xi \in \mathbb{R}^n,$$

for some positive constant λ and

$$|a_{ij}|^*_{C^\alpha(B_R)} + R|b_i|^*_{C^\alpha(B_R)} + R^2|c|^*_{C^\alpha(B_R)} \le \Lambda,$$

for some positive constant Λ.Suppose $u \in C^{2,\alpha}(B_R)$ satisfies

$$Lu = f \quad \text{in } B_R.$$

There exists an ε,depending only on n, α, λ and Λ,such that,if

$$\sup_{x,y\in B_R} |a_{ij}(x) - a_{ij}(y)| \le \varepsilon,$$

then

$$|u|^*_{C^{2,\alpha}(B_{R/2})} \le C\left(|u|_{L^\infty(B_R)} + R^2|f|^*_{C^\alpha(B_R)}\right),$$

where C is a positive constant depending only on n, α, λ and Λ.

Proof Without loss of generality,we assume $u \in C^{2,\alpha}(\overline{B_R})$.Otherwise,we consider B_r for any $r < R$,derive the desired estimate in $B_{r/2}$ and then let $r \longrightarrow R$.
We will prove

$$R^{2+\alpha}[D^2u]_{C^\alpha(B_{R/2})} \le C\left(|u|_{L^\infty(B_R)} + R^2|f|_{L^\infty(B_R)} + R^{2+\alpha}[f]_{C^\alpha(B_R)}\right).$$

This is because if we have the above estimate,**Lemma2.1(4)** implies that

$$\begin{aligned}|u|^*_{C^{2,\alpha}(B_{R/2})} &= \sum_{i=0}^2 \left(\frac{R}{2}\right)^i |D^iu|_{L^\infty(B_{R/2})} + \left(\frac{R}{2}\right)^{2+\alpha}[D^2u]_{C^\alpha(B_{R/2})}\\ &\le C_1(|u|_{L^\infty(B_{R/2})} + R|Du|_{L^\infty(B_{R/2})} + R^2|D^2u|_{L^\infty(B_{R/2})} + R^{2+\alpha}[D^2u]_{C^\alpha(B_{R/2})})\\ &\le C_1(1+C_2)(R^{2+\alpha}[D^2u]_{C^\alpha(B_{R/2})} + |u|_{L^\infty(B_{R/2})})\\ &\le C_1(1+C_2)(C_3(|u|_{L^\infty(B_R)} + R^2|f|_{L^\infty(B_R)} + R^{2+\alpha}[f]_{C^\alpha(B_R)}) + |u|_{L^\infty(B_{R/2})})\end{aligned}$$

$$
\begin{aligned}
&\leq C(|u|_{L^\infty(B_R)} + R^2|f|_{L^\infty(B_R)} + R^{2+\alpha}[f]_{C^\alpha(B_R)}) \\
&= C(|u|_{L^\infty(B_R)} + R^2|f|^*_{C^\alpha(B_R)}).
\end{aligned}
$$

Take any $B_r \subset B_R$,$r \in (0, R]$.Then,

$$
a_{ij}(0)D_{ij}u = \tilde{f} = f - cu - b_iD_iu - (a_{ij} - a_{ij}(0))D_{ij}u.
$$

Applying **Lemma2.2** in B_r,we have

$$
r^{2+\alpha}[D^2u]_{C^\alpha(B_{r/2})} \leq C\left(|u|_{L^\infty(B_r)} + r^2|\tilde{f}|_{L^\infty(B_r)} + r^{2+\alpha}[\tilde{f}]_{C^\alpha(B_r)}\right).
$$

For norms of $\tilde{f}$,we have

$$
\begin{aligned}
&r^2|\tilde{f}|_{L^\infty(B_r)} \\
\leq& r^2|f|_{L^\infty(B_r)} + r^2|c|_{L^\infty(B_r)}|u|_{L^\infty(B_r)} + r^2|b_i|_{L^\infty(B_r)}|Du|_{L^\infty(B_r)} + r^2|a_{ij} - a_{ij}(0)|_{L^\infty(B_r)}|D^2u|_{L^\infty(B_r)},
\end{aligned}
$$

and

$$
\begin{aligned}
r^{2+\alpha}[\tilde{f}]_{C^\alpha(B_r)} \leq& r^{2+\alpha}[f]_{C^\alpha(B_r)} + r^{2+\alpha}[c]_{C^\alpha(B_r)}|u|_{L^\infty(B_r)} + r^{2+\alpha}|c|_{L^\infty(B_r)}[u]_{C^\alpha(B_r)} \\
&+ r^{2+\alpha}[b_i]_{C^\alpha(B_r)}|Du|_{L^\infty(B_r)} + r^{2+\alpha}|b_i|_{L^\infty(B_r)}[Du]_{C^\alpha(B_r)} \\
&+ r^{2+\alpha}|a_{ij} - a_{ij}(0)|_{L^\infty(B_r)}[D^2u]_{C^\alpha(B_r)} + r^{2+\alpha}[a_{ij} - a_{ij}(0)]_{C^\alpha(B_r)}|D^2u|_{L^\infty(B_r)}.
\end{aligned}
$$

In fact,

$$
\begin{aligned}
[cu]_{C^\alpha(B_r)} &= \sup_{x,y\in B_r} \frac{|c(x)u(x) - c(y)u(y)|}{|x-y|^\alpha} \\
&= \sup_{x,y\in B_r} \frac{c(x)u(x) - c(x)u(y) + c(x)u(y) - c(y)u(y)}{|x-y|^\alpha} \\
&\leq \sup_{x,y\in B_r} \frac{|c(x)(u(x) - u(y))| + |u(y)(c(x) - c(y))|}{|x-y|^\alpha} \\
&\leq \sup_{x,y\in B_r} |c(x)|\frac{|u(x) - u(y)|}{|x-y|^\alpha} + \sup_{x,y\in B_r} |u(y)|\frac{|c(x) - c(y)|}{|x-y|^\alpha} \\
&\leq |c|_{L^\infty(B_r)}[u]_{C^\alpha(B_r)} + |u|_{L^\infty(B_r)}[c]_{C^\alpha(B_r)}.
\end{aligned}
$$

By the assumption on coefficients,we have

$$
|a_{ij}|_{L^\infty(B_r)} + r^\alpha[a_{ij}]_{C^\alpha(B_r)} + r|b_i|_{L^\infty(B_r)} + r^{1+\alpha}[b_i]_{C^\alpha(B_r)} + r^2|c|_{L^\infty(B_r)} + r^{2+\alpha}[c]_{C^\alpha(B_r)} \leq \Lambda,
$$

and hence,

$$
\begin{aligned}
r^{2+\alpha}[D^2u]_{C^\alpha(B_{r/2})} \leq& C(|u|_{L^\infty(B_r)} + r^2|f|_{L^\infty(B_r)} + r^{2+\alpha}[f]_{C^\alpha(B_r)} \\
&+ r^2(|a_{ij}|_{L^\infty(B_r)} + r^\alpha[a_{ij}]_{C^\alpha(B_r)})|D^2u|_{L^\infty(B_r)} \\
&+ r(r|b_i|_{L^\infty(B_r)} + r^{1+\alpha}[b_i]_{C^\alpha(B_r)})|Du|_{L^\infty(B_r)} \\
&+ (r^2|c|_{L^\infty(B_r)} + r^{2+\alpha}[c]_{C^\alpha(B_r)})|u|_{L^\infty(B_r)} \\
&+ r^\alpha[u]_{C^\alpha(B_r)} + r^{1+\alpha}[Du]_{C^\alpha(B_r)} + r^{2+\alpha}|a_{ij} - a_{ij}(0)|_{L^\infty(B_r)}[D^2u]_{C^\alpha(B_r)}) \\
\leq& C(r^2|f|_{L^\infty(B_r)} + r^{2+\alpha}[f]_{C^\alpha(B_r)} + |u|_{L^\infty(B_r)} + r^\alpha[u]_{C^\alpha(B_r)} \\
&+ r|Du|_{L^\infty(B_r)} + r^{1+\alpha}[Du]_{C^\alpha(B_r)} + r^2|D^2u|_{L^\infty(B_r)}
\end{aligned}
$$

$$+ |a_{ij} - a_{ij}(0)|_{L^\infty(B_r)} r^{2+\alpha}[D^2u]_{C^\alpha(B_r)}).$$

By the assumption of Lemma and **Lemma2.1**,we get

$$r^{2+\alpha}[D^2u]_{C^\alpha(B_{r/2})} \leq C(r^2|f|_{L^\infty(B_r)} + r^{2+\alpha}[f]_{C^\alpha(B_r)} + \varepsilon r^{2+\alpha}[D^2u]_{C^\alpha(B_r)} + C_\varepsilon|u|_{L^\infty(B_r)}).$$

In fact,this holds for any $B_r(x_0) \subset B_R$,namely,

$$r^{2+\alpha}[D^2u]_{C^\alpha(B_{r/2}(x_0))} \leq C(R^2|f|_{L^\infty(B_R)} + R^{2+\alpha}[f]_{C^\alpha(B_R)}) + C\varepsilon r^{2+\alpha}[D^2u]_{C^\alpha(B_r(x_0))} + C_\varepsilon|u|_{L^\infty(B_R)}.$$

We point out that different balls $B_{r/2}(x_0)$ and $B_r(x_0)$ are on two sides of the inequality.Set

$$Q = \sup\left\{r^{2+\alpha}[D^2u]_{C^\alpha(B_{r/2}(x_0))} \mid B_r(x_0) \subset B_R\right\}.$$

Then,for any $B_r(x_0) \subset B_R$,

$$r^{2+\alpha}[D^2u]_{C^\alpha(B_{r/4}(x_0))} \leq C\left(R^2|f|_{L^\infty(B_R)} + R^{2+\alpha}[f]_{C^\alpha(B_R)}\right) + C\varepsilon Q + C_\varepsilon|u|_{L^\infty(B_R)}.$$

We now fix a $B_r(x_0) \subset B_R$.Then,there exist $z_1, z_2, \cdots, z_N$,with $N = N(n)$,such that

$$B_{r/2}(x_0) \subset \bigcup_{i=1}^{N} B_{r/16}(z_i),$$

and

$$B_{r/2}(z_i) \subset B_R \quad for \quad i = 1, 2, \cdots, N.$$

Claim

$$[D^2u]_{C^\alpha(B_{r/2}(x_0))} \leq C\sum_{i=1}^{N}[D^2u]_{C^\alpha(B_{r/8}(z_i))}.$$

We will prove it.Take any $x_1, x_2 \in B_{r/2}(x_0)$.We assume $x_1 \in B_{r/16}(z_1)$.
If $x_2 \in B_{r/8}(z_1)$,then

$$\frac{|D^2u(x_1) - D^2u(x_2)|}{|x_1 - x_2|^\alpha} \leq [D^2u]_{C^\alpha(B_{r/8}(z_1))}.$$

If $x_2 \notin B_{r/8}(z_1)$,then $|x_1 - x_2| > \dfrac{r}{16}$.Assume $x_2 \in B_{r/8}(z_2)$.Then

$$|D^2u(x_1) - D^2u(x_2)| \leq |D^2u(x_1) - D^2u(z_1)| + |D^2u(z_1) - D^2u(z_2)| + |D^2u(z_2) - D^2u(x_2)|.$$

For z_1 and z_2,we can find $z_{k_0} = z_1, z_{k_1}, z_{k_2}, \cdots, z_{k_l} = z_2$ with $|z_{k_j} - z_{k_{j+1}}| < \dfrac{r}{8}$.We obtain

$$|D^2u(x_1) - D^2u(x_2)| \leq \left(\frac{r}{8}\right)^\alpha \sum_i [D^2u]_{C^\alpha(B_{r/8}(z_i))}.$$

This ends the proof of the claim.
For each ball $B_{r/2}(z_i) \subset B_R$,we have

$$r^{2+\alpha}[D^2u]_{C^\alpha(B_{r/4(z_i)})} \leq Q.$$

Hence,

$$r^{2+\alpha}[D^2u]_{C^\alpha(B_{r/8}(z_i))} \leq C(R^2|f|_{L^\infty(B_R)} + R^{2+\alpha}|f|_{C^\alpha(B_R)}) + C\varepsilon Q + C_\varepsilon|u|_{L^\infty(B_R)}.$$

By summing over $i=1,\cdots,N$,we have

$$\begin{aligned} r^{2+\alpha}[D^2u]_{C^\alpha(B_{r/2}(x_0))} &\le C\sum_{i=1}^{N} r^{2+\alpha}[D^2u]_{C^\alpha(B_{r/8}(z_i))} \\ &\le CN\left(R^2|f|_{L^\infty(B_R)}+R^{2+\alpha}[f]_{C^\alpha(B_R)}\right)+CN\varepsilon Q+C_\varepsilon N|u|_{L^\infty(B_R)}. \end{aligned}$$

By taking supremum over all $B_r(x_0)\subset B_R$ and ε small such that $CN\varepsilon<\dfrac{1}{2}$,we obtain

$$Q\le CN\left(R^2|f|_{L^\infty(B_R)}+R^{2+\alpha}[f]_{C^\alpha(B_R)}+|u|_{L^\infty(B_R)}\right).$$

By noting

$$R^{2+\alpha}[D^2u]_{C^\alpha(B_{R/2})}\le Q.$$

Then,we obtain the result that

$$R^{2+\alpha}[D^2u]_{C^\alpha(B_{R/2})}\le C\left(|u|_{L^\infty(B_R)}+R^2|f|_{L^\infty(B_R)}+R^{2+\alpha}[f]_{C^\alpha(B_R)}\right).$$

□

Theorem 2.1 *Let $\Omega=B_R$ be a ball in $\mathbb{R}^n$ and L be an elliptic operator in B_R.Suppose a_{ij},b_i and c are C^α-functions in B_R,for some $\alpha\in(0,1)$,and satisfy*

$$a_{ij}\xi_i\xi_j\ge\lambda|\xi|^2\quad\forall x\in B_R,\xi\in\mathbb{R}^n,$$

for some positive constant λ and

$$|a_{ij}|^*_{C^\alpha(B_R)}+R|b_i|^*_{C^\alpha(B_R)}+R^2|c|^*_{C^\alpha(B_R)}\le\Lambda,$$

for some positive constant Λ.Suppose $u\in C^{2,\alpha}(B_R)$ satisfies

$$Lu=f\quad in\ B_R.$$

Then,

$$|u|^*_{C^{2,\alpha}(B_{R/2})}\le C\left(|u|_{L^\infty(B_R)}+R^2|f|^*_{C^\alpha(B_R)}\right),$$

where C is a positive constant depending only on n,α,λ and Λ.

Proof For any $B_{R_0}(x_0)\subset B_R$,then

$$\sup_{x,y\in B_{R_0}(x_0)}|a_{ij}(x)-a_{ij}(y)|\le R_0^\alpha[a_{ij}]_{C^\alpha(B_{R_0}(x_0))}\le\left(\frac{R_0}{R}\right)^\alpha R^\alpha[a_{ij}]_{C^\alpha(B_R)}\le\left(\frac{R_0}{R}\right)^\alpha\Lambda.$$

Choose $R_0<R$ such that

$$\left(\frac{R_0}{R}\right)^\alpha\Lambda=\varepsilon.$$

By applying **Lemma2.3** in $B_{R_0}(x_0)$,we obtain

$$\begin{aligned} &R_0^{2+\alpha}[D^2u]_{C^\alpha(B_{R_0/2}(x_0))} \\ \le& C\left(R_0^2|f|_{L^\infty(B_R(x_0))}+R_0^{2+\alpha}[f]_{C^\alpha(B_R(x_0))}+|u|_{L^\infty(B_{R_0}(x_0))}\right) \\ \le& C\left(R^2|f|_{L^\infty(B_R)}+R^{2+\alpha}[f]_{C^\alpha(B_R)}+|u|_{L^\infty(B_R)}\right). \end{aligned}$$

We now take any $x, y \in B_{R/2}$.
If $|x-y| \le R_0$,by taking $x_0 = \dfrac{x+y}{2}$,we have

$$R_0^{2+\alpha}\frac{|D^2u(x)-D^2u(y)|}{|x-y|^\alpha} \le C\left(R^2|f|_{L^\infty(B_R)} + R^{2+\alpha}[f]_{C^\alpha(B_R)} + |u|_{L^\infty(B_R)}\right).$$

If $|x-y| \ge R_0$,then

$$R_0^{2+\alpha}\frac{|D^2u(x)-D^2u(y)|}{|x-y|^\alpha} \le 2R_0^2|D^2u|_{L^\infty(B_{R/2})}.$$

Hence,

$$R_0^{2+\alpha}[D^2u]_{C^\alpha(B_{R/2})} \le C\left(R^2|f|_{L^\infty(B_R)} + R^{2+\alpha}[f]_{C^\alpha(B_R)} + |u|_{L^\infty(B_R)}\right) + 2R_0^2|D^2u|_{L^\infty(B_{R/2})}.$$

Since $\dfrac{R}{R_0}$ is fixed,we obtain

$$R^{2+\alpha}[D^2u]_{C^\alpha(B_{R/2})} \le C\left(R^2|f|_{L^\infty(B_R)} + R^{2+\alpha}[f]_{C^\alpha(B_R)} + |u|_{L^\infty(B_R)} + R^2|D^2u|_{L^\infty(B_{R/2})}\right).$$

By applying **Interpolation Inequality2.1(4)** in $B_{R/2}$,we have,for any $\delta > 0$,

$$R^2|D^2u|_{L^\infty(B_{R/2})} \le \delta R^{2+\alpha}[D^2u]_{C^\alpha(B_{R/2})} + C_\delta|u|_{L^\infty(B_R)}.$$

By taking δ sufficient small,we obtain

$$R^{2+\alpha}[D^2u]_{C^\alpha(B_{R/2})} \le C\left(R^2|f|_{L^\infty(B_R)} + R^{2+\alpha}[f]_{C^\alpha(B_R)} + |u|_{L^\infty(B_R)}\right).$$

This is the desired estimate. □

• **General case.**

Theorem 2.2 (Interior $C^{2,\alpha}$ Schauder Estimates) *Let Ω be a bounded domain in $\mathbb{R}^n$ and L be an elliptic operator in Ω.Assume a_{ij},b_i and c are C^α-functions in Ω,for some $\alpha \in (0,1)$,and satisfy*

$$a_{ij}\xi_i\xi_j \ge \lambda|\xi|^2 \quad \forall x \in \Omega, \xi \in \mathbb{R}^n,$$

for some positive constant λ and

$$|a_{ij}|_{C^\alpha(\Omega)} + |b_i|_{C^\alpha(\Omega)} + |c|_{C^\alpha(\Omega)} \le \Lambda,$$

for some positive constant Λ.Suppose $u \in L^\infty(\Omega) \cap C^{2,\alpha}(\Omega)$ is a solution of

$$Lu = f \quad \text{in } \Omega,$$

for some $f \in C^\alpha(\Omega)$ with $|f|_{C^\alpha(\Omega)} < \infty$.Then,for any $\Omega' \subset\subset \Omega$,

$$|u|_{C^{2,\alpha}(\Omega')} \le C(|u|_{L^\infty(\Omega)} + |f|_{C^\alpha(\Omega)}),$$

where C is a positive constant depending only on $n, \alpha, \lambda, \Lambda, \Omega'$ and $\operatorname{dist}(\Omega', \partial\Omega)$.

Proof Set $d = \mathrm{dist}(\Omega', \partial\Omega)$.We will prove

$$d \sup_{\Omega'} |Du| + d^2 \sup_{\Omega'} |D^2 u| + d^{2+\alpha}[D^2 u]_{C^\alpha(\Omega')} \leq C \left(\sup_{\Omega} |u| + d^2 \sup_{\Omega} |f| + d^{2+\alpha}[f]_{C^\alpha(\Omega)} \right).$$

To prove this,we take any $x \in \Omega'$.Then $B_d(x) \subset \Omega$.By applying **Theorem2.1** in $B_d(x)$,we have

$$\begin{aligned} & d|Du(x)| + d^2|D^2u(x)| + d^{2+\alpha} \sup_{y \in B_{d/2}(x)} \frac{|D^2u(x) - D^2u(y)|}{|x-y|^\alpha} \\ \leq & C \left(\sup_{B_d(x)} |u| + d^2 \sup_{B_d(x)} |f| + d^{2+\alpha}[f]_{C^\alpha(B_d(x))} \right) \\ \leq & C \left(\sup_{\Omega} |u| + d^2 \sup_{\Omega} |f| + d^{2+\alpha}[f]_{C^\alpha(\Omega)} \right). \end{aligned}$$

Take any other $y \in \Omega'$.
If $|y - x| \geq \dfrac{d}{2}$,then

$$d^{2+\alpha} \frac{|D^2u(x) - D^2u(y)|}{|x-y|^\alpha} \leq 2d^2(|D^2u(x)| + |D^2u(y)|) \leq 4d^2 \sup_{\Omega'} |D^2u|.$$

The desired result follows easily. □

Corollary 2.1 *Let Ω be a bounded domain in $\mathbb{R}^n$ and L be an elliptic operator in Ω.Assume a_{ij},b_i and c are C^α-functions in Ω,for some $\alpha \in (0,1)$,and satisfies*

$$a_{ij}\xi_i\xi_j \geq \lambda|\xi|^2,$$

for some positive constant λ and

$$|a_{ij}|_{C^\alpha(\Omega)} + |b_i|_{C^\alpha(\Omega)} + |c|_{C^\alpha(\Omega)} \leq \Lambda,$$

for some positive constant Λ. Suppose $f \in C^\alpha(\Omega)$,with $|f|_{C^\alpha(\Omega)} < \infty$,and $u_k \in L^\infty(\Omega) \cap C^{2,\alpha}(\Omega)$ satisfies

$$Lu_k = f \quad \text{in } \Omega,$$

for $k = 1, 2, \cdots$,and

$$\sup_{k \geq 1} |u_k|_{L^\infty(\Omega)} < \infty.$$

Then there exist a subsequence $u_{k'}$ and a $u \in C^{2,\alpha}(\Omega)$ such that $u_{k'} \longrightarrow u$ in C^2 in any compact subset of Ω and $Lu = f$ in Ω.

2.2 Schauder estimates in weighted Hölder norms

We derive Schauder estimates in weighted Hölder norms. Let Ω be a bounded domain in $\mathbb{R}^n$.For any $x, y \in \Omega$,define

$$d_x = \mathrm{dist}(x, \partial\Omega), \quad d_{x,y} = \min\{d_x, d_y\}.$$

For a real number σ and a nonnegative integer k,we define

$$[f]_{C^k_{(\sigma)}(\Omega)} = \sup_{x \in \Omega} d_x^{k+\sigma} |D^k f(x)|,$$

$$
\begin{aligned}
|f|_{C^k_{(\sigma)}(\Omega)} &= \sum_{j=0}^{k}[f]_{C^j_{(\sigma)}(\Omega)} = \sum_{j=0}^{k}\sup_{x\in\Omega} d_x^{j+\sigma}|D^j f(x)|,\\
[f]_{C^{k,\alpha}_{(\sigma)}(\Omega)} &= \sup_{x,y\in\Omega} d_{x,y}^{k+\alpha+\sigma}\frac{|D^k f(x)-D^k f(y)|}{|x-y|^\alpha},\\
|f|_{C^{k,\alpha}_{(\sigma)}(\Omega)} &= |f|_{C^k_{(\sigma)}(\Omega)} + [f]_{C^{k,\alpha}_{(\sigma)}(\Omega)},
\end{aligned}
$$

where $\alpha \in (0,1)$.

PROPOSITION 2.1 *(1)For $\sigma+\tau \geq 0$ and $\alpha \in (0,1)$,we have*

$$|f\cdot g|_{C^\alpha_{(\sigma+\tau)}(\Omega)} \leq |f|_{C^\alpha_{(\sigma)}(\Omega)}|g|_{C^\alpha_{(\tau)}(\Omega)}$$

(2)Set

$$C^{k,\alpha}_{(\sigma)}(\Omega) = \{u \in C^{k,\alpha}(\Omega) \mid |u|_{C^{k,\alpha}_{(\sigma)}(\Omega)} < \infty\}$$

then,$C^{k,\alpha}_{(\sigma)}$ is a Banach space.

Proof
(1)

$$
\begin{aligned}
|f\cdot g|_{C^\alpha_{(\sigma+\tau)}(\Omega)} =& |f\cdot g|_{C^0_{(\sigma+\tau)}(\Omega)} + [f\cdot g]_{C^\alpha_{(\sigma+\tau)}(\Omega)}\\
=& \sup_{x\in\Omega} d_x^{\sigma+\tau}|f\cdot g(x)| + \sup_{x,y\in\Omega,x\neq y} d_{x,y}^{\alpha+\sigma+\tau}\frac{|f\cdot g(x)-f\cdot g(y)|}{|x-y|^\alpha}\\
\leq& \sup_{x\in\Omega} d_x^{\sigma}|f(x)|\cdot\sup_{x\in\Omega} d_x^{\tau}|g(x)| + \sup_{x,y\in\Omega,x\neq y} d_x^{\tau}|g(x)|\cdot d_{x,y}^{\alpha+\sigma}\frac{|f(x)-f(y)|}{|x-y|^\alpha}\\
&+ \sup_{x,y\in\Omega,x\neq y} d_y^{\sigma}|f(y)|\cdot d_{x,y}^{\alpha+\tau}\frac{|g(x)-g(y)|}{|x-y|^\alpha}\\
\leq& \left(\sup_{x\in\Omega} d_x^{\sigma}|f(x)| + \sup d_{x,y}^{\alpha+\sigma}\frac{|f(x)-f(y)|}{|x-y|^\alpha}\right)\left(\sup_{x\in\Omega} d_x^{\tau}|g(x)| + \sup d_{x,y}^{\alpha+\tau}\frac{|g(x)-g(y)|}{|x-y|^\alpha}\right)\\
=& |f|_{C^\alpha_{(\sigma)}(\Omega)}\cdot|g|_{C^\alpha_{(\tau)}(\Omega)}.
\end{aligned}
$$

(2)$C^{k,\alpha}_{(\sigma)}(\Omega)$ is a normed linear space.Choose $\{u_k\}$ is a Cauchy sequence in $C^{k,\alpha}_{(\sigma)}(\Omega)$,then exists $u \in C^k(\Omega)$,such that $u_k \longrightarrow u$ uniformly in the compact subset of $C^k(\Omega)$.Since $u_k \in C^{k,\alpha}_{(\sigma)}(\Omega)$,then $u \in C^{k,\alpha}_{(\sigma)}(\Omega)$(**Arezla-Ascolli Thoerem**). □

Theorem 2.3 *Let Ω be a bounded domain in $\mathbb{R}^n$ and L be an elliptic operator in Ω.Assume a_{ij},b_i and c are C^α-functions in Ω,for some $\alpha \in (0,1)$,and satisfy*

$$a_{ij}\xi_i\xi_j \geq \lambda|\xi|^2 \quad \forall x \in \Omega, \xi \in \mathbb{R}^n,$$

for some positive constant λ and

$$|a_{ij}|_{C^\alpha_{(0)}(\Omega)} + |b_i|_{C^\alpha_{(1)}(\Omega)} + |c|_{C^\alpha_{(2)}(\Omega)} \leq \Lambda,$$

for some positive constant Λ.Suppose $u \in C^{2,\alpha}(\Omega)$ is a solution of

$$Lu = f \quad in\ \Omega,$$

with $|u|_{C_{(-\beta)}(\Omega)} < \infty$ *and* $|f|_{C^{\alpha}_{(2-\beta)}(\Omega)} < \infty$ *for some* $\beta \in \mathbb{R}$*. Then,*

$$|u|_{C^{2,\alpha}_{(-\beta)}(\Omega)} \leq C\left(|u|_{C_{(-\beta)}(\Omega)} + |f|_{C^{\alpha}_{(2-\beta)}(\Omega)}\right),$$

where C *is a positive constant depending only on* $n, \alpha, \lambda, \Lambda$ *and* β*.*

Proof For any $x \in \Omega$,set $d_x = \mathrm{dist}(x, \partial\Omega)$ and $d = \dfrac{d_x}{2}$.By applying **Theorem2.1** in $B = B_d(x)$,we have

$$d|Du(x)| + d^2|D^2u(x)| \leq C\left(\sup_{y\in B}|u(y)| + d^2\sup_{y\in B}|f(y)| + d^{2+\alpha}\sup_{y,z\in B}\frac{|f(y)-f(z)|}{|y-z|^{\alpha}}\right).$$

Note $d \leq d_y \leq 3d$ for any $y \in B$. Notice that

$$\begin{aligned} d\cdot d_y^{-\beta} &\geq d\cdot(3d)^{-\beta} = 3^{-\beta}\cdot d^{1-\beta} \\ &= 3^{-\beta}\cdot\left(\frac{d_x}{2}\right)^{1-\beta} = 3^{-\beta}2^{\beta-1}d_x^{1-\beta} \\ \Longrightarrow & d_x^{1-\beta} \leq C\cdot d\cdot d_y^{-\beta}. \end{aligned}$$

Then

$$\begin{aligned} & d_x^{1-\beta}|Du(x)| + d_x^{2-\beta}|D^2u(x)| \\ \leq & C\left(\sup_{y\in B} d_y^{-\beta}|u(y)| + \sup_{y\in B} d_y^{2-\beta}|f(y)| + \sup_{y,z\in B} d_{y,z}^{2+\alpha-\beta}\frac{|f(y)-f(z)|}{|y-z|^{\alpha}}\right) \\ \leq & C\left(|u|_{C_{(-\beta)}(\Omega)} + |f|_{C^{\alpha}_{(2-\beta)}(\Omega)}\right). \end{aligned}$$

Hence,

$$|u|_{C^{2}_{(-\beta)}(\Omega)} \leq C\left(|u|_{C_{(-\beta)}(\Omega)} + |f|_{C^{\alpha}_{(2-\beta)}(\Omega)}\right).$$

To estimate $[u]_{C^{2,\alpha}_{(-\beta)}(\Omega)}$,let x, y be distinct points in Ω with $d_x \leq d_y$ and $B = B_d(x)$ as above.We have,for $|x-y| \leq \dfrac{d}{2}$,apply **Theorem2.1**

$$\begin{aligned} d_{x,y}^{2+\alpha-\beta}\frac{|D^2u(x)-D^2u(y)|}{|x-y|^{\alpha}} &= (2d)^{2+\alpha-\beta}\cdot\frac{|D^2u(x)-D^2u(y)|}{|x-y|^{\alpha}} \\ &\leq C2^{2+\alpha-\beta}\cdot d^{-\beta}(|u|_{L^{\infty}(B)} + d^2|f|^*_{C^{\alpha}(B)}) \\ &= C(d^{-\beta}|u|_{L^{\infty}(B)} + d^{2-\beta}|f|^*_{C^{\alpha}(B)}) \\ &= C(|u|_{C_{(-\beta)}(B)} + |f|_{C^{\alpha}_{(2-\beta)}(B)}) \\ &\leq C(|u|_{C_{-\beta}(\Omega)} + |f|_{C^{\alpha}_{(2-\beta)}(\Omega)}), \end{aligned}$$

and,for $|x-y| > \dfrac{d}{2}$,

$$\begin{aligned} d_{x,y}^{2+\alpha-\beta}\frac{|D^2u(x)-D^2u(y)|}{|x-y|^{\alpha}} &\leq d_x^{2+\alpha-\beta}\frac{|D^2u(x)|+|D^2u(y)|}{(d/2)^{\alpha}} \\ &= 4^{\alpha}d_x^{2-\beta}(|D^2u(x)|+|D^2u(y)|) \\ &\leq 8[u]_{C^{2}_{(-\beta)}(\Omega)}. \end{aligned}$$

In conclusion,

$$d_{x,y}^{2+\alpha-\beta}\frac{|D^2u(x)-D^2u(y)|}{|x-y|^\alpha}\le C(|u|_{C_{(-\beta)}(\Omega)}+|f|_{C^\alpha_{(2-\beta)}(\Omega)})+8[u]_{C^2_{(-\beta)}(\Omega)}.$$

Taking the supremum with respect to x,y,we have

$$\begin{aligned}[u]_{C^{2,\alpha}_{(-\beta)}(\Omega)}&\le C(|u|_{C_{(-\beta)}(\Omega)}+|f|_{C^\alpha_{(2-\beta)}(\Omega)}+[u]_{C^2_{(-\beta)}(\Omega)})\\&\le C(|u|_{C_{(-\beta)}(\Omega)}+|f|_{C^\alpha_{(2-\beta)}(\Omega)}).\end{aligned}$$

because we have the estimate about $|u|_{C^2_{-\beta}(\Omega)}$.Then,we have the desired estimate. □

Corollary 2.2 *Let $\Omega=B$ be a ball in $\mathbb{R}^n$ and L be an elliptic operator in B.Assume a_{ij},b_i and c are C^α-functions in B,for some $\alpha\in(0,1)$,and satisfy*

$$a_{ij}\xi_i\xi_j\ge\lambda|\xi|^2,$$

$c\le 0$ and

$$|a_{ij}|_{C^\alpha_{(0)}(B)}+|b_i|_{C^\alpha_{(1)}(B)}+|c|_{C^\alpha_{(2)}(B)}\le\Lambda,$$

for some positive constant Λ.Suppose $u\in C^{2,\alpha}(B)$ is a solution of

$$\begin{cases}Lu=f & \text{in } B\\ u=0 & \text{on } \partial B\end{cases}$$

with $|f|_{C^\alpha_{(2-\beta)}(B)}<\infty$ for some $\beta\in(0,1)$.Then,

$$|u|_{C^{2,\alpha}_{(-\beta)}(B)}\le C|f|_{C^\alpha_{(2-\beta)}(B)},$$

where C is a positive constant depending only on n,α,λ,Λ and β.

Proof By **Lemma3.3.4**,we have

$$|u|_{C_{(-\beta)}(B)}\le C|f|_{C_{(2-\beta)}(B)}.$$

Hence, the desired result follows from **Theorem2.3**. □

2.3 The Dirichlet problem–methods of continuity(balls)

We discuss the Dirichlet problem with continuous boundary values for general uniformly elliptic equations.

• Balls(homogeneous boundary values)—Method of continuity.

Theorem 2.4 *Let $\Omega=B$ be a ball in $\mathbb{R}^n$ and L be an elliptic operator in B.Assume a_{ij},b_i and c are C^α-functions in B,for some $\alpha\in(0,1)$,and satisfy*

$$a_{ij}\xi_i\xi_j\ge\lambda|\xi|^2\quad\forall x\in\Omega,\xi\in\mathbb{R}^n,$$

for some positive λ,$c\le 0$ and

$$|a_{ij}|_{C^\alpha_{(0)}(B)}+|b_i|_{C^\alpha_{(1)}(B)}+|c|_{C^\alpha_{(2)}(B)}\le\Lambda,$$

for some constant Λ.Then,for any $f\in C^\alpha(B)$ with $|f|_{C^\alpha_{(2-\beta)}}<\infty$ for some $\beta\in(0,1)$,there exists a solution

$u \in C(\bar{B}) \cap C^{2,\alpha}(B)$ *of the Dirichlet problem*

$$\begin{cases} Lu = f & in \quad B \\ u = 0 & on \quad \partial B. \end{cases}$$

Moreover

$$|u|_{C^{2,\alpha}_{(-\beta)}(B)} \leq C|f|_{C^{\alpha}_{(2-\beta)}(B)},$$

where C is a positive constant depending only on $n, \alpha, \lambda, \Lambda$ and β.

Proof We consider the family of equations

$$L_t u \equiv tLu + (1-t)\Delta u = f,$$

for $t \in [0,1]$.We note that $L_0 = \Delta$ and $L_1 = L$.
Set

$$L_t u = a^t_{ij} D_{ij} u + b^t_i D_i u + c^t u,$$

where

$$\begin{cases} a^t_{ij} = ta_{ij} + (1-t)\delta_{ij} \\ b^t_i = tb_i \\ c^t = tc. \end{cases}$$

We have

$$\begin{aligned} a^t_{ij}\xi_i\xi_j &= (ta_{ij} + (1-t)\delta_{ij})\xi_i\xi_j \\ &= ta_{ij}\xi_i\xi_j + (1-t)|\xi|^2 \\ &\geq t\lambda|\xi|^2 + (1-t)|\xi|^2 \\ &= (t(\lambda-1)+1)|\xi|^2 \geq \min\{1,\lambda\}|\xi|^2, \end{aligned}$$

for any $x \in B$ and $\xi \in \mathbb{R}^n$,and

$$\begin{aligned} &|a^t_{ij}|_{C^{\alpha}_{(0)}(B)} + |b^t_i|_{C^{\alpha}_{(1)}(B)} + |c^t|_{C^{\alpha}_{(2)}(B)} \\ =&|ta_{ij} + (1-t)\delta_{ij}|_{C^{\alpha}_{(0)}(B)} + |tb_{ij}|_{C^{\alpha}_{(1)}(B)} + |tc|_{C^{\alpha}_{(2)}(B)} \\ \leq& t(|a_{ij}|_{C^{\alpha}_{(0)}(B)} + |b_i|_{C^{\alpha}_{(1)}(B)} + |c|_{C^{\alpha}_{(2)}(B)}) + (1-t)n \\ \leq& t\Lambda + (1-t)n \leq \max\{n,\Lambda\}, \end{aligned}$$

independent of $t \in [0,1]$.By direct calculation,we have

$$\begin{aligned} |L_t u|_{C^{\alpha}_{(2-\beta)}(B)} &\leq \sum_{i,j=1}^{n} |a^t_{ij} D_{ij} u|_{C^{\alpha}_{(2-\beta)}(B)} + \sum_{i=1}^{n} |b^t_i D_i u|_{C^{\alpha}_{(2-\beta)}(B)} + |c^t u|_{C^{\alpha}_{(2-\beta)}(B)} \\ &\leq \sum_{i,j=1}^{n} |a^t_{ij}|_{C^{\alpha}_{(0)}(B)} |D_{ij} u|_{C^{\alpha}_{(2-\beta)}(B)} + \sum_{i=1}^{n} |b^t_i|_{C^{\alpha}_{(1)}(B)} |D_i u|_{C^{\alpha}_{(1-\beta)}(B)} + |c^t|_{C^{\alpha}_{(2)}(B)} |u|_{C^{\alpha}_{(-\beta)}(B)} \\ &\leq C \cdot |u|_{C^{2,\alpha}_{(-\beta)}(B)} \textbf{(Why?)}, \end{aligned}$$

where C is a positive constant depending only on $n, \alpha, \lambda, \Lambda, \beta$ and $\mathrm{diam}(B)$.Then,$L_t : C^{2,\alpha}_{(-\beta)}(B) \longrightarrow C^{\alpha}_{(2-\beta)}(B)$ is a bounded linear operator,for each $t \in [0,1]$.

We now let I be the collection of $s \in [0,1]$ such that the Dirichlet problem

$$\begin{cases} L_s u = f & \text{in } B \\ u = 0 & \text{on } \partial B \end{cases}$$

is solvable in $C^{2,\alpha}_{(-\beta)}(B)$ for any $f \in C^{\alpha}_{(2-\beta)}(B)$.
For $s = 0$,$L_s = \Delta$.By **Theorem1.6** and **Corollary2.2**,the Dirichlet problem is solvable,then $0 \in I$.
We take $s \in I$ and let $u = L_s^{-1} f$ be the solution.**Corollary2.2** implies

$$|L_s^{-1} f|_{C^{2,\alpha}_{(-\beta)}} \leq C|f|_{C^{\alpha}_{(2-\beta)}(B)}.$$

For any $t \in [0,1]$ and $f \in C^{\alpha}_{(2-\beta)}(B)$,we write $L_t u = f$ as

$$\begin{aligned} L_s u &= f + (L_s - L_t)u \\ &= f + (sLu + (1-s)\Delta u - tLu - (1-t)\Delta u) \\ &= f + (s-t)Lu + \Delta u - s\Delta u - \Delta u + t\Delta u \\ &= f + (t-s)(\Delta u - Lu). \end{aligned}$$

Hence,$u \in C^{2,\alpha}_{(-\beta)}(B)$ is a solution of

$$\begin{cases} L_t u = f & \text{in } B \\ u = 0 & \text{on } \partial B \end{cases}$$

if and only if

$$u = L_s^{-1}(f + (t-s)(\Delta u - Lu)).$$

For any $u \in C^{2,\alpha}_{(-\beta)}(B)$,set

$$Tu = L_s^{-1}(f + (t-s)(\Delta u - Lu)).$$

Then $T : C^{2,\alpha}_{(-\beta)}(B) \longrightarrow C^{2,\alpha}_{(-\beta)}(B)$ is an operator and,for any $u, v \in C^{2,\alpha}_{(-\beta)}(B)$,

$$\begin{aligned} |Tu - Tv|_{C^{2,\alpha}_{(-\beta)}(B)} &= |(t-s)L_s^{-1}((\Delta - L)(u-v))|_{C^{2,\alpha}_{(-\beta)}(B)} \\ &\leq C|t-s||(\Delta - L)(u-v)|_{C^{\alpha}_{(2-\beta)}(B)} \\ &\leq C|t-s||u-v|_{C^{2,\alpha}_{(-\beta)}(B)}. \end{aligned}$$

Therefore,$T : C^{2,\alpha}_{(-\beta)}(B) \longrightarrow C^{2,\alpha}_{(-\beta)}(B)$ is a contraction if

$$|t-s| < \delta = C^{-1}.$$

Hence for any $t \in [0,1]$ with $|t-s| < \delta$,there exists $u \in C^{2,\alpha}_{(-\beta)}(B)$ such that $u = Tu$,that is,

$$u = L_s^{-1}(f + (t-s)(\Delta u - Lu)).$$

In other words,for any $t \in [0,1]$ with $|t-s| < \delta$ and any $f \in C^{\alpha}_{(2-\beta)}(B)$,there exists a solution $u \in C^{2,\alpha}_{(-\beta)}(B)$ of

$$\begin{cases} L_t u = f & \text{in } B \\ u = 0 & \text{on } \partial B. \end{cases}$$

Hence,if $s \in I$,then $t \in I$ for any $t \in [0,1]$ with $|t-s| < \delta$.We now divide the interval $[0,1]$ into subintervals of length less that δ.By $0 \in I$,we conclude $1 \in I$,that is,

$$\begin{cases} Lu = f & \text{in } B \\ u = 0 & \text{on } \partial B. \end{cases}$$

□

• **Balls(nonhomogeneous boundary values).**

Theorem 2.5 *Let $\Omega = B$ be a ball in $\mathbb{R}^n$ and L be an elliptic operator in B.Assume a_{ij},b_i and c are C^α-functions in B,for some $\alpha \in (0,1)$,with finite C^α-norms in B,and satisfy*

$$a_{ij}\xi_i\xi_j \geq \lambda|\xi|^2 \quad \forall x \in B, \xi \in \mathbb{R}^n,$$

and $c \leq 0$.Then,for any $f \in L^\infty(B) \cap C^\alpha(B)$ and any $\varphi \in C(\partial B)$,there exists a solution $u \in C(\bar{B}) \cap C^{2,\alpha}(B)$ of the Dirichlet problem

$$\begin{cases} Lu = f & \text{in } B \\ u = \varphi & \text{on } \partial B. \end{cases}$$

Proof We assume that φ is defined as a continuous function in $\bar{B}$.Let $\{\varphi_k\}$ be a sequence of functions in $C^3(\bar{B})$ converging uniformly to φ in $\bar{B}$.By **Theorem2.4**,there exists a unique solution $v_k \in C(\bar{B}) \cap C^{2,\alpha}(B)$ of

$$\begin{cases} Lv_k = f - L\varphi_k & \text{in } B \\ v_k = 0 & \text{on } \partial B. \end{cases}$$

Define $u_k = v_k + \varphi_k$.Then $u_k \in C(\bar{B}) \cap C^{2,\alpha}(B)$ and satisfies

$$\begin{cases} Lu_k = f & \text{in } B \\ u_k = \varphi_k & \text{on } \partial B. \end{cases}$$

For any k and l,

$$\begin{cases} L(u_k - u_l) = 0 & \text{in } B \\ u_k - u_l = \varphi_k - \varphi_l & \text{on } \partial B. \end{cases}$$

The maximum principle implies

$$\sup_B |u_k - u_l| \leq \sup_{\partial B} |\varphi_k - \varphi_l| \leq \sup_{\partial B} |\varphi_k - \varphi| + \sup_{\partial B} |\varphi_l - \varphi| \longrightarrow 0.$$

Therefore,$\{u_k\}$ converges uniformly in $\bar{B}$ to a function $u \in C(\bar{B})$ such that $u = \varphi$ on ∂B.By the compactness provided by **Corollary2.1**,we have $u \in C^{2,\alpha}(B)$ and $Lu = f$ in B.Hence,u is the desired solution. □

2.4 The Dirichlet problem–Perron's method(general bounded domain)

We now discuss the Dirichlet problem in general bounded domains by Perron's method.Essentially used are the maximum principle and the solvability of the Dirichlet problems in balls.
We first discuss subsolutions.
A C^2-function v is a subsolution if $Lv \geq f$.We now introduce notions of subsolutions and supersolutions for continuous functions.

Lemma 2.4 *Let Ω be a domain in $\mathbb{R}^n$ and L be an elliptic operator in Ω.Assume a_{ij},b_i and c are C^α-functions in Ω,for some $\alpha \in (0,1)$,and satisfy*

$$a_{ij}\xi_i\xi_j \geq \lambda|\xi|^2 \quad \forall x \in \Omega, \xi \in \mathbb{R}^n,$$

and $c \leq 0$.Suppose $v \in C^2(\Omega)$ and $f \in C^\alpha(\Omega)$.Then,$Lv \geq f$ in Ω if and only if,for any ball $B \subset\subset \Omega$ and any $w \in C(\bar{B}) \cap C^2(B)$ with $Lw = f$,$v \leq w$ on ∂B implies $v \leq w$ in B.

Proof
"$\Longrightarrow$".For any ball $B \subset \Omega$ and any $w \in C(\bar{B}) \cap C^2(B)$ with $Lw = f$ in B and $v \leq w$ on ∂B,we have

$$\begin{cases} L(v-w) \geq 0 & \text{in } B \\ v - w \leq 0 & \text{on } \partial B. \end{cases}$$

By the maximum principle ,we have $v \leq w$ in B.
"$\Longleftarrow$".If $Lv < f$ somewhere in Ω,that is,exists $x_0 \in \Omega$ such that $Lv(x_0) < f(x_0)$.Then exists $B_\varepsilon(x_0) \subset\subset \Omega$,such that $Lv(x) < f(x)$ in $B_\varepsilon(x_0)$.Let $w \in C(\bar{B}_\varepsilon(x_0)) \cap C^{2,\alpha}(B_\varepsilon(x_0))$ solve

$$\begin{cases} Lw = f & \text{in } B_\varepsilon(x_0) \\ w = v & \text{on } \partial B_\varepsilon(x_0). \end{cases}$$

The existence of w is implied by **Theorem2.5**.We have $v \leq w$ in $B_\varepsilon(x_0)$ by the assumption.Next we note that

$$\begin{cases} L(v-w) < 0 & \text{in } B_\varepsilon(x_0) \\ v - w = 0 & \text{on } \partial B_\varepsilon(x_0). \end{cases}$$

We have $w \leq v$ in $B_\varepsilon(x_0)$ by the maximum principle.Hence $v = w$ in $B_\varepsilon(x_0)$,which contradicts $Lw > Lv$ in $B_\varepsilon(x_0)$.Therefore,$Lv \geq f$ in Ω. □

Definition 2.1 (Subsolution,Supersolution) *Let Ω be a domain in $\mathbb{R}^n$ and L be an elliptic operator in B.Assume a_{ij},b_i and c are continuous functions in Ω and satisfy*

$$a_{ij}\xi_i\xi_j \geq \lambda|\xi|^2 \quad \forall x \in \Omega, \xi \in \mathbb{R}^n,$$

*and $c \leq 0$.Suppose f is a continuous function in Ω.A function $v \in C(\Omega)$ is a **subsolution**(or **supersolution**) in Ω if for any ball $B \subset\subset \Omega$ and any solution $w \in C^2(B)$ of $Lw = f$ in B,$v \leq (\geq)$ w on ∂B implies $v \leq (\geq)$ w in B.*

Remark 2.1 *The above definition of subsolutions (supersolution) are defined for continuous functions.*

Lemma 2.5 *Let Ω be a domain in $\mathbb{R}^n$ and L be an elliptic operator in Ω.Assume a_{ij},b_i and c are C^α-functions in Ω,for some $\alpha \in (0,1)$,and satisfy*

$$a_{ij}\xi_i\xi_j \geq \lambda|\xi|^2 \quad \forall x \in \Omega, \xi \in \mathbb{R}^n,$$

and $c \leq 0$.Suppose $u, v \in C(\bar{\Omega})$ and $f \in C^\alpha(\Omega)$.If u is subsolution in Ω and v is supersolution in Ω with $u \leq v$ on $\partial\Omega$,then $u \leq v$ in Ω.

Proof We first note that $u - v \leq 0$ on $\partial\Omega$.Set $M = \max\limits_{\bar{\Omega}}(u - v)$ and

$$D = \{x \in \Omega \mid u(x) - v(x) = M\} \subset \Omega.$$

We consider only the case $M \geq 0$.Then D is relatively closed by the continuity of u and v.
Next we prove that D is open.
For any $x_0 \in D$,we take $r < \mathrm{dist}(x_0, \partial\Omega)$.Let $\bar{u}, \bar{v} \in C(\bar{B}) \cap C^{2,\alpha}(B)$ solve,respectively,

$$\begin{cases} L\bar{u} = f & \text{in } B_r(x_0) \\ \bar{u} = u & \text{on } \partial B_r(x_0) \end{cases}$$

and

$$\begin{cases} L\bar{v} = f & \text{in } B_r(x_0) \\ \bar{v} = v & \text{on } \partial B_r(x_0). \end{cases}$$

The existence of $\bar{u}$ and $\bar{v}$ is implied by **Theorem2.5**.Definition implies

$$u \leq \bar{u}, \quad \bar{v} \leq v \quad \text{in } B_r(x_0).$$

Hence,

$$\bar{u} - \bar{v} \geq u - v \quad \text{in } B_r(x_0).$$

Next,

$$\begin{cases} L(\bar{u} - \bar{v}) = 0 & \text{in } B_r(x_0) \\ \bar{u} - \bar{v} = u - v & \text{on } \partial B_r(x_0). \end{cases}$$

With $u - v \leq M$ on $\partial B_r(x_0)$,the maximum principle implies $\bar{u} - \bar{v} \leq M$ in $B_r(x_0)$.In particular,

$$M \geq (\bar{u} - \bar{v})(x_0) \geq (u - v)(x_0) = M.$$

Hence,$(\bar{u}-\bar{v})(x_0) = M$ and then $\bar{u}-\bar{v}$ has an interior maximum at x_0.By the strong maximum principle,$\bar{u}-\bar{v} \equiv M$ in $\bar{B}_r(x_0)$.
Therefore,$u - v = M$ on $\partial B_r(x_0)$.This holds for any $r < \mathrm{dist}(x_0, \partial\Omega)$.Then,$u - v = M$ in $B_r(x_0)$ and hence $B_r(x_0) \subset D$.
In conclusion,D is both relatively closed and open in Ω.Therefore,either $D = \emptyset$ or $D = \Omega$.
In other words,$u - v$ either attains its nonnegative maximum only on $\partial\Omega$ or $u - v$ is constant.Since $u \leq v$ in $\partial\Omega$,we have $u \leq v$ in Ω in both cases. □

Now we demonstrate how to generate greater subsolutions from given subsolutions.

Lemma 2.6 *Let Ω be a domain in $\mathbb{R}^n$ and L be an elliptic operator in Ω.Assume a_{ij},b_i and c are C^α-functions in Ω,for some $\alpha \in (0,1)$,and satisfy*

$$a_{ij}\xi_i\xi_j \geq \lambda|\xi|^2 \quad \forall x \in \Omega, \xi \in \mathbb{R}^n,$$

and $c \leq 0$.Suppose $f \in C^\alpha(\Omega)$.Assume $v \in C(\bar{\Omega})$ is a subsolution in Ω and B is a ball in Ω with $\bar{B} \subset \Omega$.Let w be defined by

$$w = v \quad \text{in } \bar{\Omega} \setminus B$$

and

$$Lw = f \quad \text{in } B.$$

Then w is a subsolution in Ω and $v \leq w$ in $\bar{\Omega}$.

Proof The existence of w in B is implied by **Theorem2.5**.Then,$w \in C(\bar{B}) \cap C^{2,\alpha}(B)$.We also have $v \leq w$

in B by definition of subsolution.
Next,we take any B' with $\bar{B}' \subset \Omega$ and consider $u \in C(\bar{B}') \cap C^{2,\alpha}(B')$ with $Lu = f$ in B' and $w \leq u$ on $\partial B'$.By $v \leq w$ on $\partial B'$,we have $v \leq u$ on $\partial B'$.
Then,v is a subsolution and u is a solution in B' with $v \leq u$ on $\partial B'$.By **Lemma2.5**,we have $v \leq u$ in B'.Hence $w \leq u$ in $B' \setminus B$(by $w = v$ in $\bar{\Omega} \setminus B$).
Next,$Lw = Lu$ in $B \cap B'$ and $w \leq u$ on $\partial(B \cap B')$.By the maximum principle,we have $w \leq u$ in $B \cap B'$.Hence $w \leq u$ in B'.
Therefore,w is a subsolution in Ω by Definition. □

Remark 2.2 *The function w is called the **lifting** of v by the operator L in B.*

Remark 2.3 *Lemma asserts that we obtain greater subsolution if we preserve the values of subsolutions outside the balls and extend them inside the balls by solutions.*

Now we are ready to solve the Dirichlet problem by Perron's method.

• Creating a solution in Ω related to the boundary value φ.

Lemma 2.7 *Let Ω be a bounded domain in $\mathbb{R}^n$ and L be an elliptic operator in Ω.Assume a_{ij},b_i and c are C^α-functions in Ω,for some $\alpha \in (0,1)$,and satisfy*

$$a_{ij}\xi_i\xi_j \geq \lambda|\xi|^2 \quad \forall x \in \Omega, \xi \in \mathbb{R}^n,$$

and $c \leq 0$.Suppose f is a bounded C^α-function in Ω and φ is a continuous function on $\partial\Omega$.Define,for any $x \in \Omega$,

$$u_\varphi(x) = \sup\{v \mid v \in C(\bar{\Omega}) \quad \textit{is a subsolution in } \Omega \textit{ and } v \leq \varphi \textit{ on } \partial\Omega\}.$$

Then,$u_\varphi \in C^{2,\alpha}(\Omega)$ and $Lu_\varphi = f$ in Ω.

Proof Set

$$\mathcal{S}_\varphi = \{v \mid v \in C(\bar{\Omega}) \quad \text{is a subsolution in } \Omega \text{ and } v \leq \varphi \text{ on } \partial\Omega\}.$$

Then,for any $x \in \Omega$,

$$u_\varphi = \sup\{v(x) \mid v \in \mathcal{S}_\varphi\}.$$

In the following,we simply write $\mathcal{S} = \mathcal{S}_\varphi$.
Step 1. *We prove that u_φ is well defined.*
We assume Ω lies in $\{0 < x_1 < d\}$ and set

$$F = \sup_\Omega |f|, \qquad \Phi = \max_{\partial\Omega} |\varphi|.$$

Define

$$v_+(x) = \Phi + (e^{\gamma d} - e^{\gamma x_1})F,$$

$$v_-(x) = -\Phi - (e^{\gamma d} - e^{\gamma x_1})F.$$

A straightforward calculation yields

$$Lv_+ = -(a_{11}\gamma^2 + b_1\gamma)Fe^{\gamma x_1} + cv_+.$$

We choose γ large enough so that

$$a_{11}\gamma^2 + b_1\gamma \geq 1 \quad \text{in } \Omega.$$

Since $e^{\gamma x_1} \geq 1$,$c \leq 0$ and $v_+ \geq 0$ in Ω,we have

$$Lv_+ \leq -F \leq f \quad \text{in } \Omega.$$

The boundary condition $v_+ \geq \varphi$ on $\partial\Omega$ is trivial.
A similar argument works for v_-.Then

$$\begin{cases} Lv_+ \leq f & \text{in } \Omega \\ v_+ \geq \varphi & \text{on } \partial\Omega \end{cases}$$

and

$$\begin{cases} Lv_- \geq f & \text{in } \Omega \\ v_- \leq \varphi & \text{on } \partial\Omega. \end{cases}$$

By the definition of $\mathcal{S}$,we have $v_- \in \mathcal{S}$.Hence,the set $\mathcal{S}$ is not empty.
By **Lemma2.5**,for any $v \in \mathcal{S}$,

$$v \leq v_+ \quad \text{in } \Omega.$$

Therefore,u_φ is well-defined and $u_\varphi \leq v_+$ in Ω.
Step 2. *We prove that $\mathcal{S}$ is closed by taking maximum among finitely many functions in $\mathcal{S}$.*
We take arbitrary $v_1, v_2, \cdots, v_k \in \mathcal{S}$ and set

$$v = \max\{v_1, \cdots, v_k\}.$$

We take any ball $B \subset \Omega$ and any $w \in C(\bar{B}) \cap C^2(B)$ with $Lw = f$ in B and $v \leq w$ on ∂B.Then $v_i \leq w$ on ∂B,for $i = 1, 2, \cdots, k$.Since v_i is aubsolution,we get $v_i \leq w$ in B,so $v \leq w$ in B.We conclude that v is a subsolution in Ω.Hence $v \in \mathcal{S}$.
Step 3. *For any $B_r(x_0) \subset \Omega$,we prove that $u_\varphi \in C^{2,\alpha}(B_r(x_0))$ and $Lu = f$ in $B_r(x_0)$.*
By the definition of u_φ,there exists a sequence of functions $v_i \in \mathcal{S}$ such that

$$\lim_{i\to\infty} v_i(x_0) = u_\varphi(x_0).$$

We may replace v_i above by any $\tilde{v}_i \in \mathcal{S}$ with $\tilde{v}_i \geq v_i$,since

$$v_i(x_0) \leq \tilde{v}_i(x_0) \leq u_\varphi(x_0).$$

Replacing,if necessary,v_i by $\max\{v_-, v_i\} \in \mathcal{S}$,we may assume

$$v_- \leq v_i \leq u_\varphi \quad \text{in } \Omega.$$

For the fixed $B_r(x_0)$ and each v_i,we let w_i be the lifting of v_i,that is,$w_i = v_i$ in $\Omega \setminus B_r(x_0)$ and $Lw_i = f$ in $B_r(x_0)$.By **Lemma2.6**,$w_i \in \mathcal{S}$(by $w_i = v_i$ in $\Omega \setminus B_r(x_0)$) and $v_i \leq w_i$ in Ω.Hence,

$$\lim_{i\to\infty} w_i(x_0) = u_\varphi(x_0),$$

and

$$v_- \leq w_i \leq u_\varphi \quad \text{in } \Omega,$$

for any $i = 1, 2, \cdots$.
Then,$\{w_i\}$ is a bounded sequence satisfying $Lw_i = f$ in $B_r(x_0)$.
By **Corollary2.1**,there exists a function $w \in C^{2,\alpha}(B_r(x_0))$ such that a subsequence of $\{w_i\}$,still denoted by

$\{w_i\}$,converges to w in any compact subset of $B_r(x_0)$ and $Lw = f$ in $B_r(x_0)$.Then,

$$w \leq u_\varphi \text{ in } B_r(x_0) \text{ and } w(x_0) = u_\varphi(x_0).$$

We now claim $u_\varphi = w$ in $B_r(x_0)$.
If exists $z \in B_r(x_0)$,such that $w(z) < u_\varphi(z)$.By the definition of u_φ,there exists $\bar{v} \in \mathcal{S}$ such that $w(z) < \bar{v}(z)$. Set $u_i = \max\{\bar{v}, w_i\} \in \mathcal{S}$ and let U_i be the lifting of u_i.Then $U_i \in \mathcal{S}$ and $\{U_i\}$ is a bounded sequence satisfying $LU_i = f$ in $B_r(x_0)$.
By **Corollary2.1**,there exists a function $U \in C^{2,\alpha}(B_r(x_0))$ such that a subsequence of $\{U_i\}$,still denoted by $\{U_i\}$,converges to U in any compact subset of $B_r(x_0)$ and $LU = f$ in $B_r(x_0)$.Then,

$$w \leq U \leq u_\varphi \text{ in } B_r(x_0) \text{ and } w(x_0) = U(x_0) = u_\varphi(x_0).$$

Since $Lw = f = LU$ in $B_r(x_0)$ and $w(x_0) - U(x_0) = 0 = \max\limits_{B_r(x_0)}(w(x) - U(x))$,thus

$$w \equiv U \quad \text{in } B_r(x_0).$$

Since $U(z) \geq \bar{v}(z) > w(z)$,then $u_\varphi = w$ in $B_r(x_0)$.Therefore,$u_\varphi \in C^{2,\alpha}(B_r(x_0))$ and $Lu_\varphi = f$ in $B_r(x_0)$. $\square$

• **Discussing limits of u_φ as x approaches the boundary.**

Lemma 2.8 *Let Ω be a bounded domain in $\mathbb{R}^n$ and L be an elliptic operator in Ω.Assume a_{ij}, b_i and c are C^α-functions in Ω,for some $\alpha \in (0,1)$,and satisfy*

$$a_{ij}\xi_i\xi_j \geq \lambda|\xi|^2 \quad \forall x \in \Omega, \xi \in \mathbb{R}^n,$$

and $c \leq 0$.Suppose f is a bounded C^α-function in Ω and φ is a continuous function on $\partial\Omega$.Define,for any $x \in \Omega$,

$$u_\varphi(x) = \sup\{v \mid v \in C(\bar{\Omega}) \quad \textit{is a subsolution in } \Omega \textit{ and } v \leq \varphi \textit{ on } \partial\Omega\}.$$

For some $x_0 \in \partial\Omega$,suppose $w_{x_0} \in C(\bar{\Omega}) \cap C^2(\Omega)$ satisfies

$$\begin{cases} Lw_{x_0} \leq -1 & \textit{in } \Omega \\ w_{x_0}(x_0), w_{x_0}(x) > 0 & \forall x \in \partial\Omega \setminus \{x_0\}. \end{cases}$$

Then,

$$\lim_{x \to x_0} u_\varphi(x) = \varphi(x_0).$$

Proof We set

$$\mathcal{S}_\varphi = \{v \mid v \in C(\bar{\Omega}) \quad \text{is a subsolution in } \Omega \text{ and } v \leq \varphi \text{ on } \partial\Omega\}.$$

We simply write $w = w_{x_0}$ and set $M = \max\limits_{\partial\Omega}|\varphi|$.
Let ε be an arbitrary positive constant.By the continuity of φ at x_0,there exists a positive constant δ such that

$$|\varphi(x) - \varphi(x_0)| \leq \varepsilon,$$

for any $x \in \partial\Omega \cap B_\delta(x_0)$.We then choose K_ε sufficiently large so that

$$K_\varepsilon w(x) \geq 2M \geq |\varphi(x) - \varphi(x_0)|,$$

for any $x \in \partial\Omega \setminus B_\delta(x_0)$.Hence,

$$|\varphi(x) - \varphi(x_0)| \leq \varepsilon + K_\varepsilon w \quad \text{on } \partial\Omega.$$

Next,taking $\tilde{K}_\varepsilon = \max\{K_\varepsilon, \sup_\Omega |f - c\varphi(x_0)|\}(-\tilde{K}_\varepsilon \leq f - c\varphi(x_0) \leq \tilde{K}_\varepsilon)$,we have

$$\begin{aligned} L(\varphi(x_0) + \varepsilon + \tilde{K}_\varepsilon w) &= c\varphi(x_0) + c\varepsilon + \tilde{K}_\varepsilon Lw \\ &\leq c\varphi(x_0) - \tilde{K}_\varepsilon \\ &\leq f \quad \text{in } \Omega \end{aligned}$$

and

$$\begin{aligned} L(\varepsilon(x_0) - \varepsilon - \tilde{K}_\varepsilon w) &= c\varphi(x_0) - c\varepsilon - \tilde{K}_\varepsilon Lw \\ &\geq c\varphi(x_0) + \tilde{K}_\varepsilon \\ &\geq f \quad \text{in } \Omega. \end{aligned}$$

Since $\varepsilon(x_0) - \varepsilon - \tilde{K}_\varepsilon w(x)$ is a subsolution in Ω with $\varphi(x_0) - \varepsilon - \tilde{K}_\varepsilon w \leq \varphi$ on $\partial\Omega$,we have $\varphi(x_0) - \varepsilon - \tilde{K}_\varepsilon w \in \mathcal{S}_\varepsilon$.The definition of u_φ implies

$$\varphi(x_0) - \varepsilon - \tilde{K}_\varepsilon w \leq u_\varphi \quad \text{in } \Omega.$$

On the other hand,$\varphi(x_0) + \varepsilon + \tilde{K}_\varepsilon w$ is a supersolution in Ω with $\varphi(x_0) + \varepsilon + \tilde{K}_\varepsilon w \geq \varphi(x)$ on $\partial\Omega$.Hence,for any $v \in \mathcal{S}_\varphi$,we obtain,by **Lemma2.5**,

$$v \leq \varphi(x_0) + \varepsilon + \tilde{K}_\varepsilon w \quad \text{in } \Omega.$$

Again by the definition of u_φ,we have

$$u_\varphi \leq \varphi(x_0) + \varepsilon + \tilde{K}_\varepsilon w \quad \text{in } \Omega.$$

Therefore,

$$|u_\varphi(x) - \varphi(x_0)| \leq \varepsilon + \tilde{K}_\varepsilon w \quad \text{in } \Omega.$$

This implies

$$\limsup_{x \to x_0} |u(x) - \varphi(x_0)| \leq \varepsilon.$$

We obtain the desired result by letting $\varepsilon \longrightarrow 0$. □

Remark 2.4 *The function w_{x_0} is called a **barrier function**.*

Theorem 2.6 *Let Ω be a bounded domain in $\mathbb{R}^n$ satisfying an exterior sphere condition at each of its boundary point and L be an elliptic operator in Ω.Assume a_{ij}, b_i and c are C^α-functions in Ω with bounded C^α-norms in Ω,for some $\alpha \in (0,1)$,and satisfy*

$$a_{ij}\xi_i\xi_j \geq \lambda|\xi|^2 \quad \forall x \in \Omega, \xi \in \mathbb{R}^n,$$

and $c \leq 0$.Then for any $f \in L^\infty(\Omega) \cap C^\alpha(\Omega)$ and $\varphi \in C(\partial\Omega)$,there exists a (unique) solution $u \in C(\bar{\Omega}) \cap C^{2,\alpha}(\Omega)$ of the Dirichlet problem

$$\begin{cases} Lu = f & \text{in } \Omega \\ u = \varphi & \text{on } \partial\Omega. \end{cases}$$

Proof Combining **Lemma2.7**,**Lemma2.8** and **Lemma3.3.2**,we obtain the desired result. □

Remark 2.5 *In summary,Perron's method yields a solution of the Dirichlet problem for uniformly elliptic*

euqations.This method depends essentially on the maximum principle and the solvability of the Dirichlet problem in balls.An important feature here is that the interior existence problem is separated from the boundary behavior of solutions,which is determined by the local geometry of domains.

Index

$C^{k,\alpha}$-boundary, 5

Barrier Function, 39

Freeze Coefficients, 22

Hölder Continuous, 5
Hölder Norm, 6
Hölder Semi-norm, 6
Hölder Space, 6

Interior $C^{2,\alpha}$ Schauder Estimates, 26
Interpolation Inequality, 18

Lifting, 36
Locally Hölder Continuous, 5

Method of Continuity, 30

Newtonian Potential, 6

Poisson Equation, 1

Strictly Elliptic, 17
Subsolution, 34
Supersolution, 34